MSc thesis project

Research proposal and student guide

Trapped-ion quantum technology • UV fibre delivery at 369.5 nm

Experimental demonstration of reduced fibre solarisation in an oxygen-free atmosphere

Can a 3D-printed controlled-atmosphere enclosure stop the UV-induced degradation of single-mode fibre end faces carrying milliwatts of 369.5 nm light?

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Project at a glance

- **Problem.** Thorlabs polarisation-maintaining UV fibres carrying a few mW of 369.5 nm light (the $^{171}Yb^{+}$ cooling and detection wavelength) couple very well after a fresh polish, then lose throughput at a rate that grows with power, at both ends.
- **Idea.** Keep the end faces in an oxygen-free atmosphere, using 3D-printed cells with oxygen absorbers or an inert-gas purge, and show that the degradation stops.
- **Approach.** A multi-arm accelerated-ageing testbed runs ambient air, clean dry air, dry N_2 and sealed absorber cells *in parallel* from one UV source. A fixed cleaning sequence and surface analysis locate the damage. The best solution is then demonstrated on a working 369.5 nm link.
- **Why it is a good project.** Every outcome is useful. Either oxygen is the culprit and the fix is cheap, or the experiment identifies the real cause (contamination, water, particles or intrinsic glass damage) and what to do about it.

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Part I – Research proposal

1 Summary

Trapped-ion experiments with $^{171}Yb^{+}$ deliver their 369.5 nm cooling and detection light through single-mode, polarisation-maintaining optical fibres. In the group, these fibres show a characteristic failure. Directly after the connector end faces are polished, the coupling efficiency is excellent. It then falls over days to weeks, faster at higher power, and both the input and the output end are affected. Re-polishing restores the throughput, so the damage must sit at, or within micrometres of, the end faces rather than in the bulk of the fibre. At a few milliwatts in a mode-field diameter of about 2.4 µm, the end faces see of order 10^5 W/cm^2 of near-UV light, an intensity at which photochemistry of the glass surface and of anything adsorbed on it becomes important.

This project tests the hypothesis that **removing oxygen from the end faces halts the degradation**, and develops a practical way of doing so with **3D-printed enclosures** that are either purged with inert gas or sealed with an **oxygen absorber**. The literature does not settle the question in advance. Hydrocarbons are photo-fixed into absorbing deposits on UV-illuminated optics, and against this, oxygen and often water vapour are consistently *protective*. Space lasers, UV frequency-conversion cavities and laser diodes are operated in oxygen-containing atmospheres for exactly this reason. Siloxanes released by silicones behave differently: oxygen converts them into non-volatile SiO_x, and at a 410 nm fibre end face oxygen has been found to *accelerate* such deposits. This scenario supports the group's hypothesis. Airborne particles and a restructuring of the silica surface itself are further candidates. The experiment is therefore designed to separate these factors. A multi-arm testbed exposes identical fibres *simultaneously* to ambient air, clean dry synthetic air, clean dry nitrogen and sealed absorber cells, and later to controlled oxygen, humidity and contaminant levels. Throughput, back-reflection and the atmosphere in every cell are logged continuously. At the end of each run a fixed cleaning and re-polishing sequence, together with surface analysis, shows where the loss is and what it consists of.

The project runs over 12 months (a 6-month variant is included). It starts with a forensic study of fibres already degraded in the lab, builds and qualifies the testbed and cells, runs two atmosphere campaigns, and ends with a demonstration on a working 369.5 nm link. Success means a quantitative answer to whether oxygen matters (with confidence intervals), and a qualified cell design that holds $< 100\,\mathrm{ppm}\ O_2$ for months. If the atmosphere is the key, the degradation should fall below 1 % per 1000 h at 3 mW. Part II of this document is a practical guide for the student: onboarding plan, core skills, working practices, pitfalls, a prioritised reading list and draft procedures.

2 Background and motivation

2.1 Near-UV light delivery in trapped-ion experiments

In $^{171}Yb^{+}$ experiments, light at 369.5 nm drives the $^2S_{1/2}$–$^2P_{1/2}$ transition used for Doppler cooling, optical pumping and state detection. Light at 399 nm is used for isotope-selective photoionisation of neutral Yb. Delivering this light through single-mode fibres has clear advantages. The fibre acts as a spatial filter that produces a clean Gaussian mode. It decouples the laser table from the vacuum system and keeps beam pointing at the ion stable. With polarisation-maintaining (PM) fibre, it also delivers a well-defined polarisation. Scalable architectures with many ion-trap modules depend even more on fibre delivery, so the long-term reliability of near-UV fibres is a practical concern for the whole field.

2.2 The problem in the group

The group uses Thorlabs PM patch cables designed for the near UV (referred to in the lab as "375PM"; Section 3.3) at powers of a few milliwatts. The behaviour observed is:

- directly after the end faces have been polished, the coupling efficiency is excellent;
- it then declines over time, faster when more power is used;
- both the input and the output end are affected;
- polishing the end faces again restores the throughput.

The practical consequences are frequent re-polishing (person-hours and risk of damaging the connector), slow drifts of the power delivered to the ions, the need for spare laser power, and occasionally sudden failures during measurements.

What these observations already tell us. Re-polishing removes only micrometres to tens of micrometres of glass. Colour centres formed along the length of the fibre would survive it. The restoration therefore places the damage at, or within micrometres of, the end faces. The power dependence shows that the process is driven by light. The degradation of the *output* face, which sees almost the same guided intensity as the input face but no focusing optics, argues against simple misalignment of the input coupling and for a process driven by local intensity at a glass–gas interface. At 3 mW in a 2.4 µm mode-field diameter, the peak intensity is $\approx 1.3 \times 10^5$ W/cm^2, and each square centimetre of the core region receives about 2.5×10^{23} photons per second (Appendix A). At such photon fluxes even very weak absorption by surface sites or adsorbed molecules leads to photochemistry within seconds.

2.3 Terminology and scope

In the lab, "solarisation" is used for any UV-induced loss of fibre throughput. Strictly, solarisation means the formation of colour centres in the glass (Section 3.1). In this document, *end-face degradation* denotes the observed phenomenon and *solarisation* denotes colour-centre formation. Finding out which of the two is happening, or whether both are, is part of the project (RQ1). The scope is the atmosphere at the end faces of the fibre type used in the group, at 369.5 nm and a nearby testbed wavelength (375 nm). Developing new fibres is out of scope, but alternative fibre technologies are reviewed (Section 3.3) because they are the fall-back if the atmosphere turns out not to be the cause.

3 State of the art

3.1 UV-induced defects in silica and their relevance at 369.5 nm

Energetics. A 369.5 nm photon carries 3.36 eV; two photons carry 6.71 eV. The absorption edge of amorphous SiO_2 lies near 8 eV [1], so neither one nor two photons can excite the perfect glass across its gap. Excimer-laser studies show that E′ centres form in a two-photon process at 5.0 eV and 6.4 eV [2, 3]. XeCl light (4.0 eV, two-photon energy 8.0 eV) created no centres in dry synthetic silica [2]. At 369.5 nm only processes that start at a pre-existing defect, an impurity or a surface site are therefore expected: one-photon absorption into a weak band, or stepwise two-photon absorption through a long-lived intermediate state.

The relevant defects. Table 1 lists the main point defects of silica glass [4–8]. Most absorb well above 3.36 eV, but several channels exist at 369.5 nm:

- the spin-forbidden singlet–triplet band of the twofold-coordinated silicon centre ODC(II) at 3.15 eV, whose long-lived (ms) triplet state can absorb a second photon;
- the 6.9 eV band of ODC(II) and the onset of the Schumann–Runge absorption of interstitial O_2 (shifted to ≈ 6.5 eV in the glass), both reachable by two photons; photolysed interstitial O_2 yields mobile oxygen atoms, interstitial ozone and peroxy radicals [9–12];
- chlorine species left by the manufacturing process: interstitial Cl_2 absorbs at 3.78 eV with a width of 0.6 eV, and atomic Cl^0 at 3.26 eV. In pure-silica-core fibres, much of the radiation-induced loss above 2.5 eV is chlorine-related [8, 13].

Table 1: Main point defects in silica glass: absorption and photoluminescence (PL) bands, in eV, after [4, 8, 13, 18]. Values in bold are within reach of one (3.36 eV) or two (6.71 eV) photons at 369.5 nm.

Defect	Absorption (eV)	PL (eV)	Comment
E′ centre, ≡Si$^{\bullet}$	5.8	–	dominant UV-induced centre
NBOHC, ≡Si–O$^{\bullet}$	1.97, 4.8 (continuum 4–7.8)	1.91	3.36 eV lies in its low-absorption gap
ODC(I), ≡Si–Si≡	7.6	4.4	precursor of E′
ODC(II), =Si:	5.05, **6.9**; triplet **3.15**	4.4, 2.75	triplet band and long-lived T_1 state
peroxy radical, ≡Si–O–O$^{\bullet}$	≈2.0, 4.8	–	forms from NBOHC + O, or E′ + O_2
interstitial O_2	onset ≈**6.5**	0.975	source of O atoms and O_3
interstitial O_3	4.8	–	bleached by UV
interstitial Cl_2, Cl^0	**3.78** (FWHM 0.6), **3.26**	–	chlorine from manufacture
GeODC (Ge-doped cores only)	5.15; triplet ≈**3.8**	4.3, 3.2	Ge photosensitivity

In pure-silica-core single-mode fibre under 405 nm CW light at MW/cm^2, bleaching of ODC(II) and growth of E′ centres have indeed been observed [14]. Germanium-related centres would add a one-photon channel through the GeODC triplet band near 3.8 eV. This is the origin of the photosensitivity of Ge-doped fibres at near-UV and, via two photons, visible wavelengths [15–17]. The group's PM-S350-HP fibre has a pure-silica core, so these channels should be absent.

Hydrogen, OH and fibre manufacture. Molecular hydrogen passivates E′ centres and NBOHCs. The E′ + H_2 reaction has an activation energy of only 0.38 eV and proceeds at the same rate for surface E′ centres [19, 20]. Under ArF and F_2 irradiation H_2 can also promote ODC(I) formation [21]. Fibre drawing creates E′ centres, NBOHCs (the 630 nm band, which grows with draw tension) and peroxy radicals [22–24]. Strained Si–O–Si bonds are the main intrinsic precursors [25]. Decades of work on multimode UV fibres have led from hydrogen-loaded to hydrogen-free, high-OH designs with improved deep-UV stability [26–30]. Loading with H_2 and UV curing has transferred this to single-mode PCF [31].

Kinetics. Colour-centre growth in fibres typically follows a stretched exponential or a power law in dose [32, 33]. Two-photon creation shows up as a quadratic dependence on intensity [2, 3]. Comparable methods for measuring and fitting photodarkening have been standardised for Yb-doped fibres [34–36]. Measuring the initial rate at three or more powers therefore gives a cheap first indication of the mechanism.

Gas transport into the glass. At room temperature H_2 diffuses through a 125 µm fibre in a few days [37, 38]. Oxygen molecules move about ten orders of magnitude more slowly. Extrapolated high-temperature data [1, 39] give penetration depths of 0.1–5 nm per year (Appendix A). The bulk of the core is therefore isolated from the atmosphere, except for deliberately added hydrogen. Oxygen can act only at the surface, where it dissolves readily [40].

Reactive surfaces. Freshly fractured or ground silica carries reactive sites: silicon and oxygen dangling bonds, silylene groups and strained two-membered rings [41–44]. These react with H_2O (which opens the strained rings [45]), with H_2 and with O_2. Polished end faces are such mechanically activated surfaces. Oxygen, water and hydrogen therefore determine what UV-generated surface defects turn into: peroxy radicals, silanols or Si–H. These products absorb mainly above 4 eV, not at 3.36 eV. Spectroscopically, H1 is therefore not the most likely mechanism, which is one more reason to test it directly rather than assume it.

3.2 What has been observed in single-mode fibres carrying UV and blue light

Table 2 collects the most relevant reports. Two kinds of degradation emerge clearly. *Bulk solarisation* dominates in the deep UV. It is irreversible, it can set in at microwatt powers in fused-silica fibres [31], and what light remains is carried in a speckled mode pattern [46]. Below about 300 nm, unprotected photonic-crystal fibres lose most of their transmission within hours at milliwatt powers [31, 47]. *End-face degradation* is reversible by re-cleaving or re-polishing, and it is the more relevant kind at near-UV and blue wavelengths. It takes at least three forms:

- **Deposits.** Colombe et al. mention UV-induced carbon deposition on input faces [31]. Marciniak et al. saw deposits on large-mode-area PCF faces at 313 nm with input powers as low as 10 mW. Polishing removed them, and a slight overpressure of dry N_2 at the collimators prevented them for months [46]. They attributed the deposits to particles and oil drawn into the beam by the "optical-tweezer effect" and then burned in. Purged and unpurged ends of the same fibre behaved differently. A Duke group reports the same use of nitrogen purging against particle deposition at UV wavelengths [48].
- **Restructuring of the silica.** In the most detailed study of CW blue light in small-core single-mode fibres (405 nm, mode-field diameter 2 – 3 µm), Gonschior et al. found no deposits. Instead, laser-induced periodic surface structures formed on both faces: protrusions of 0.5 – 0.7 µm after 9–11 days, ripples of 250 – 400 nm period aligned with the polarisation, and a porous, lower-density silica layer. EDX and ToF-SIMS found no foreign elements [14, 49]. The loss grew faster than linearly in time, typically 0.2 – 1 dB/day at 74 – 95 mW. The output beam quality degraded (M^2 from 1.2 to 5 in 12 days). Cleaved faces degraded four times more slowly than polished ones [49, 50]. The structures appeared at 0.5 – 2.5 MW/cm^2, three orders of magnitude below the optical damage threshold [14].
- **Slow drifts of unclear origin.** At 369 nm, Sandia found lensed-fibre-to-chip coupling stable at 7 – 24 mW on short time scales but degrading over hours, with surface contamination of the fibre or chip facets among the suspected causes [51].

Two further observations bear directly on the oxygen hypothesis. Gonschior ran a PM single-mode fibre for 50 days at 250 mW (405 nm) between *nitrogen-purged, sealed* collimators. The fibre lost only about 1 dB, but cone-like structures still formed on both faces [49]. Inert gas alone therefore did not stop the restructuring at that (much higher) intensity. Second, in ion-trap vacuum systems, 369 – 370 nm light caused losses in oxide mirror coatings that were reversed by admitting oxygen or air [52, 53]. Absence of oxygen can therefore *harm* UV-illuminated oxide surfaces.

What is not known

We have found no published measurement of the degradation of step-index, pure-silica-core PM fibre at 369.5 nm and a few milliwatts, the configuration used in the group. No study in the trapped-ion community has treated O_2 as a controlled variable. Nitrogen purging is used to keep particles away, not to remove oxygen. This is the gap the project fills.

3.3 Fibre technologies for near-UV delivery

The group's fibre. The cables known in the group as "375PM" are Thorlabs P3-375PM-FC patch cables (1 – 10 m), FC/APC at both ends, aligned at 375 nm, with no AR-coated option [57]. They are built on PM-S350-HP: a PANDA fibre with a *pure-silica core*, mode-field diameter 2.3 µm at 350 nm (about 2.4 µm at 369.5 nm), NA 0.12 and second-mode cut-off ≤ 340 nm [58]. A germanium-free core avoids the strong photosensitivity of Ge-doped silica, and Thorlabs advertises pure-silica cores for their resistance to photodarkening. The same catalogue nevertheless states that all fibres eventually photodarken in the UV and should be treated as consumables. For CW light at an air–glass face of undoped silica, Thorlabs quotes a practical safe intensity of about 250 kW/cm^2 [59]. With Thorlabs' convention this corresponds to about 11 mW for a 2.4 µm mode. The group's few milliwatts are below that, yet the faces still degrade,

Table 2: Selected reports on UV/blue degradation of single-mode fibres and related surfaces. PCF: photonic-crystal fibre; LMA: large mode area; HC: hollow core.

Ref.	λ (nm)	Fibre / surface	Observation	Mitigation that worked
[31]	313, 280	LMA PCF	bulk solarisation within hours at mW; carbon on input face	H_2 loading + UV curing; collapsed, angled ends
[47]	280	solid-core PCF; kagomé HC	solid core $-80\,\%$ in 3 h at 2 mW; HC unchanged after 100 h at 15 mW	hollow core
[54]	250	LMA PCF	damage mainly at the input; partly reversed by re-cleaving	low power ($< 0.3\,\mathrm{mW}$)
[14, 49]	405	SM and PM, MFD 2–3 µm	LIPSS and porous silica on both faces; M^2 grows; no foreign elements	end caps or launch fibre: $< 0.03\,\mathrm{dB/day}$
[46]	313 (369)	LMA-PM PCF	particle/oil deposits from 10 mW; removed by polishing	dry N_2 overpressure
[51]	369	lensed fibre to chip	slow (hour-scale) coupling loss	–
[52, 53]	369–422	oxide mirror coatings in vacuum	UV-induced loss in vacuum	reversed by O_2 or air
[55, 56]	355	optics with organic contaminants	deposit growth in vacuum	O_2 partial pressure

which illustrates that "damage threshold" figures say little about slow photochemical processes. Thorlabs no longer sells the bare fibre; Coherent (Nufern) offers a fibre with the same part number. The 405 nm PM cables use PM-S405-XP, whose cut-off ($380 \pm 20\,\mathrm{nm}$) may allow a second mode at 369.5 nm. They are therefore no substitute.

Large-mode-area photonic-crystal fibre with hydrogen loading. Endlessly single-mode PCFs spread the light over roughly ten times the mode area. Loading them with H_2 (about 10 MPa, room temperature, 4–6 days) and curing them with UV light before the hydrogen escapes makes them resistant to deep-UV solarisation for hundreds of hours at 125 mW (313 nm) [31, 60]. Hydrogen loading alone does not confer resistance unless curing occurs before the hydrogen has diffused out [31]. Collapsing the holes over a few hundred micrometres and angle-cleaving the ends protects the faces and suppresses back-reflections. PM versions reach polarisation extinction ratios of 30–32 dB at 369 nm after curing [46]. These fibres are not off-the-shelf patch cords, and their faces still collect deposits without a purge [46].

Hollow-core fibres. Kagomé and antiresonant hollow-core fibres guide almost all the light in gas, so the glass hardly sees it. They have transmitted 280 nm CW light without change for 100 h [47] and 320 nm light at up to 60 mW without degradation, at about 0.13 dB/m [61]. Losses below 10 dB/km have been reported at 369 nm [62], and guidance extends down to 190 nm [63]. Caveats: there is no PM hollow-core fibre, ammonium chloride can grow on cleaved hollow-core end faces [64], and reversible photodarkening has been reported in UV-guiding hollow-core fibres [65].

End caps. A coreless silica cap spliced to the fibre lets the beam expand before it reaches the glass–gas interface. At 369.5 nm a 300 µm cap reduces the face intensity about 280-fold (Appendix A). At 405 nm, a short launch fibre and end caps reduced the damage rate from 0.2 – 1 dB/day to below 0.03 dB/day [49]. End caps attack the intensity-driven surface processes at their root. Together with the atmosphere control studied here they are the most promising engineering solution, and the obvious next step if the atmosphere alone is not enough.

3.4 Surface processes and the role of the atmosphere

Four communities have studied UV-driven degradation of optical surfaces in controlled atmospheres: space lidar, UV frequency-conversion cavities, semiconductor lasers and laser modules, and high-power fibre connectors. Their results are the best available guide to what the atmosphere around a fibre end face will do.

Laser-induced contamination in space lasers. A prototype of the 355 nm ALADIN laser for ESA's Aeolus mission lost 50 % of its transmission in less than six hours. Outgassed organics were carbonised on the optics by the UV light, and even a few parts per billion of organics caused contamination [66]. The flight laser therefore operates with a low-pressure oxygen purge of a few tens of pascals [66, 67]. Systematic tests at DLR showed that an O_2-to-contaminant pressure ratio above 50:1 drastically reduces deposit growth, above 1000:1 no transmission loss was detectable, and existing deposits were completely removed by irradiation in several tens of pascals of oxygen [55, 68–70]. The ESA/DLR guidance for ground operation is explicit: "operation with normal dry air is to be preferred to operation in nitrogen" [71]. The same paper notes that silicones are not removed by oxidation, that CW irradiation can produce more deposit than pulsed irradiation, and that high-purity bottled gas alone does not prevent absorbing "black dots". In sealed 1064 nm lasers, trace contamination from ultra-high-purity nitrogen cylinders was enough to cause damage above about 60 °C [72]. With 300 ppm toluene present, optics survived 8×10^3 pulses in N_2, 1.8×10^4 in synthetic air, and more than 10^6 in air with a few thousand ppm of water or alcohol [73].

CW evidence at our intensity. Parthasarathy et al. studied contamination at a diamond surface in vacuum under 1.5 mW of CW 532 nm light focused to about 220 kW/cm^2, an intensity close to that on the group's fibre faces [74]. Deposits grew *fastest* at an oxygen partial pressure of about 5×10^{-3} mbar. Between 2×10^{-3} mbar and 2 mbar their onset was delayed, and above 2 mbar none formed. Humidity slowed the growth. At atmospheric pressure, 1 – 1000 ppm of O_2 corresponds to 0.001 – 1 mbar. An "almost oxygen-free" cell is therefore exactly in the regime where contamination was promoted, whereas air is a hundred times above the inhibition threshold. How these vacuum results transfer to 1 atm of nitrogen is not known. That is one of the questions the **O2-x** arms answer.

UV conversion cavities. Groups producing CW UV light for trapped ions routinely flush their nonlinear crystals with oxygen. At NIST the 313 nm BBO crystal sits in an enclosure "purged with a gentle flow of dry oxygen", there mainly against moisture-induced fogging [75]. At PTB oxygen is led directly onto the crystal facets "to prevent light-induced damage" [76]. At 243 nm, cavity mirrors degraded within minutes below about 0.7 mbar of oxygen, and the deposit could be removed with organic solvents [77, 78]. A recent watt-level 230 nm source uses filtered dry air because "pure nitrogen removes this cleaning pathway" [79]. One industrial patent reports the opposite for BBO: clean dry air alone degraded the crystal within 100 hours, and adding about 1000 ppm of water to the purge cured this [80].

Semiconductor lasers and fibre-coupled laser modules. GaN laser diodes operated in pure nitrogen grow a thick, highly absorbing deposit on the facet, which oxygen removes [81]. Carbon deposition [82] and the growth of SiO_x layers [83, 84] are documented facet failure modes. Package patents mostly specify oxygen-containing fills: more than 100 ppm [85], at least 5 % [86], or 1 – 18 % [87]. For 405 nm lasers, oxygen is said to react with photo-decomposed silicon compounds before they deposit on the facet [88]. The important exception concerns *fibre* end faces. In a fibre-coupled 410 nm module, reliability *deteriorated* at oxygen concentrations of 100 ppm or more, because silicon-based compounds deposited at the fibre end face. Silicone parts were the principal source, and the patent specifies 1 – 100 ppm O_2 [89]. A related patent on siloxane-derived SiO_x deposits at a fibre input recommends silicone-free materials and air admitted only through activated carbon [90]. Another describes a "dust collection effect" where the optical density is high [91].

Can 369.5 nm light make ozone?. Splitting O_2 needs photons below 242 nm (the Herzberg continuum [92]). UV/ozone cleaning therefore uses the 185 nm and 254 nm mercury lines [93]. Two 369.5 nm photons carry enough energy, but no two-photon cross-section near 370 nm is known, and any effect would be confined to the ≈20 µm next to the face. Oxygen suppresses contamination even where it cannot be photodissociated [55, 74]. The relevant chemistry is light-assisted oxidation at the surface, not gas-phase ozone.

Particles and high-power connectors. Dielectric particles are trapped by the optical gradient force in air [94, 95], and absorbing particles are moved by photophoretic forces [96]. Fibre connectors carrying watts at 1550 nm fail mainly because of absorbing contamination on the face, at intensities comparable to ours [97–99]. The 313 nm fibre deposits of Marciniak et al. were attributed to this mechanism [46].

Oxygen: friend or foe?

- **Hydrocarbon contamination:** every community finds oxygen (and often water vapour) *protective*. Nitrogen and vacuum favour absorbing carbon deposits, most strongly at trace O_2 levels.
- **Siloxane contamination:** oxidation turns silicones into non-volatile SiO_x. At a 410 nm fibre end face, oxygen above 100 ppm *accelerated* such deposits. This is the strongest support in the literature for the group's hypothesis.
- **Particles and silica restructuring:** not expected to depend on oxygen. A clean, slightly pressurised gas keeps particles away.

The outcome for the group's fibres therefore depends on which contaminants are present, and finding that out is the first task (Phase 0). A sealed, nearly oxygen-free cell that contains outgassing polymers is a high-risk configuration. The experiment tests it against clean, oxygen-containing gas rather than assuming it helps.

3.5 Creating and verifying a clean, oxygen-free atmosphere

Oxygen absorbers. Iron-based absorbers (Mitsubishi Gas Chemical AGELESS) reduce the oxygen in a sealed barrier container to $\leq 0.1\,\%$. The container wall should let through no more than 20 mL of oxygen per square metre, day and atmosphere of pressure difference [100]. The number in the type name is the volume of O_2 absorbed: a ZP-50 binds 50 mL. Some types need moisture from their surroundings. The self-reacting Z types work best above about 55 % RH and release humidity into the container [101]. The reaction is exothermic, and the sachets lose capacity within minutes in air. Lambert et al. give a useful rule for a leaking box: the equilibrium concentration is $[O_2] \approx L/(12.7\,C)$, where L is the ingress in ppm per day and C is the absorber capacity divided by the box volume [102]. The RP System of the same manufacturer exists in two types. RP-A absorbs oxygen, water and corrosive gases and drives the humidity to nearly zero. RP-K leaves the humidity unchanged but can release organic acids if overloaded [103]. Both reach $\leq 0.1\,\%$ O_2 within a day and shed fine powder, so the sachet belongs behind a membrane. Colour indicators change near 0.1 – 0.5 % O_2 and show only gross failures. Vacuum non-evaporable getters need activation at 350 – 450 °C and would saturate almost immediately in a cell at atmospheric pressure [104]. In heated form they serve as in-line gas purifiers. Commercial point-of-use purifiers deliver inert gas with < 100 ppt O_2 and H_2O [105]. A palladium catalyst with a few per cent H_2 holds anaerobic chambers at 0 – 5 ppm O_2 [106], an option if forming gas is used anyway.

Measuring oxygen, water and organic vapour. Proving $< 100\,\mathrm{ppm}$ needs an instrument that resolves ppm.

- Ultra-trace luminescent sensor spots read through a window: e.g. PreSens SP-PSt9, 0 – 200 ppm, detection limit 0.5 ppm [107]. They are ideal for sealed cells. The common trace spots with a 50 ppm detection limit are only marginal for this purpose [108], and percent-level sensors are useless here [109].
- Galvanic trace analysers (detection limit ∼0.05 ppm) need a sample flow of 0.5 – 1 L/min. They suit the exhaust of purged arms rather than sealed cells [110].
- Zirconia sensors read wrongly in hydrogen-containing gas and are poisoned by silicone vapour [111].

Luminescent spots must be shielded from the UV beam. Capacitive humidity sensors resolve trends down to about 1 % RH (≈230 ppmv) [112]. Low-cost metal-oxide VOC sensors depend on oxygen adsorbed on the oxide [113], so they work only in oxygen-containing streams. Organic vapour in oxygen-free streams is best sampled onto sorbent tubes and analysed by TD-GC/MS.

Table 3: Permeability at about 23 °C (1 Barrer $= 3.35 \times 10^{-16}\,\mathrm{mol\,m\,m^{-2}\,s^{-1}\,Pa^{-1}}$) and outgassing (ASTM E595 total mass loss TML and collected volatile condensable material CVCM) of candidate cell materials. Values compiled from [119, 126, 131–135]; – not found. Permeabilities of PETG, ASA, ABS and PEI should be measured by rate of rise.

Material	O_2 (Barrer)	H_2O (Barrer)	TML / CVCM (%)	Remark
PET / PETG	0.034 (PET)	130 (PET)	0.35 / 0.02 (PETG)	good barrier; low VOC emission when printed
PLA	0.15–0.43	2000–3300	0.70–0.75 / 0.01	very water-permeable
ABS	–	≈2600	0.59–0.94 / 0.01–0.04	emits styrene; avoid
PC	1.0	–	0.14–0.17 / 0.00	
PP	1.7	–	–	poor O_2 barrier
PA12 (SLS)	0.13–0.20	–	0.75 / 0.02	hygroscopic
PEEK	0.14	≈140	0.26–0.30 / ≤0.02	needs a high-temp. printer
ULTEM 9085 / Antero 800NA	–	–	0.33–0.37 / ≤0.02; 0.35 / 0.004	low-outgassing FDM
SLA resins (Clear, Tough, Grey)	0.125 (VeroClear)	–	1.5–2.9 / ≤0.02	Rigid 10K: 0.54 / 0.00
PVDF / PCTFE	0.031 / 0.093	–	0.27 / 0.01 (PVDF)	excellent barriers
PTFE / FEP	2.3 / 4.5	–	–	poor O_2 barrier despite inertness
FKM / butyl O-rings	1.3–1.7	40–200	0.10–0.82 / ≤0.04 (FKM)	preferred seal materials
polyurethane	1.5–4.7	340–12 500	–	avoid for gas lines
silicone	100–600	10 500	–	never near the end faces

Gas purity. Nitrogen 5.0 contains up to 3 ppm O_2, 5 ppm H_2O and 0.2 ppm hydrocarbons; 6.0 grade one sixth of that [114]. "Zero grade" air may contain 5 ppm of hydrocarbons [115], whereas ultra-zero grades are specified at 0.1 ppm [116]. The hydrocarbon content must be specified explicitly, because parts-per-billion levels already matter (Appendix A).

3D-printed enclosures. FDM parts become leak-tight with slight over-extrusion (extrusion multiplier 0.98 to 1.2 depending on the printer), walls of at least 1.1 mm and randomised seams [117]. Thin layers matter more than infill [118], and laser-sintered PA12 permeates like bulk material [119]. Printed parts are used in vacuum systems: SLS PA12 was characterised by Povilus et al. [120]; FDM PEEK, SLA resins and other printed plastics have been tested since [121–123]. Printed metal chambers reach ultra-high vacuum in portable quantum technology [124]. Table 3 lists permeabilities and outgassing data. Three points stand out:

- Dense polymer walls let in 50–500 times more water molecules than oxygen molecules. Printed ABS stores 0.35 wt% of gas, mostly water [125]. A sealed "dry" cell therefore becomes wet within hours unless it contains a desiccant (Appendix A).
- Standard SLA resins lose 1.5 – 2.9 % of their mass in vacuum, whereas PETG, PEEK, ULTEM and Antero lose about 0.3 % [126]. FDM printing of ABS releases styrene, and cured resins still emit acrylates and aldehydes [127–129].
- O-rings need a surface finish of about 0.4 µm, squeeze of at most 30 % and 60 – 85 % gland fill [130]. FDM surfaces are much rougher, so the seal should sit on an ironed top skin, a machined land or an SLA part. Silicone O-rings let in about 200 times more oxygen than FKM or butyl, and they release siloxanes.

3.6 Technical facts that shape the testbed

- **Source.** The only single-mode 375 nm laser diode in Nichia's line-up is the NDU4116 (70 mW), sold by Thorlabs as L375P70MLD [136]. It needs a TEC mount that supports its pin code (e.g. LDM56F). Splitting its free-space beam before any fibre makes every input face a test item, and no upstream delivery fibre ages. With 30 – 60 % coupling efficiency, four to six arms at 3 mW guided are possible.
- **Detectors.** Silicon photodiodes lose UV responsivity under exposure [137, 138]; photodiodes with high UV resistance exist [139]. SiC photodiodes respond only weakly at 369.5 nm because it lies at their band edge.
- **Lenses.** The moulded aspheres commonly used for fibre coupling are made of high-index glasses whose UV transmission is reduced. Check them at 369.5 nm.
- **Inspection standards.** The core zone of IEC 61300-3-35 has a diameter of 25 µm, about ten times the mode-field diameter [140]. Standard pass/fail criteria are too coarse for these fibres. An in-house criterion based on $\geq 400\times$ imaging of the central 10 µm is needed.
- **Polishing.** Thorlabs' procedure for FC/APC uses diamond films down to 1 µm and a final 0.01 µm silica film, and warns that alumina and SiC films undercut fibres in zirconia ferrules [141]. Every final film and lubricant also leaves a possible residue on the face. The procedure is therefore part of the experimental protocol.

3.7 Contribution of this project

The project combines four elements that, to our knowledge, have not been brought together: (i) the step-index pure-silica PM fibre and the wavelength and power actually used in $^{171}Yb^{+}$ experiments; (ii) atmosphere as a controlled variable, with O_2, water, organic vapour and particles separated; (iii) simultaneous, identical arms with continuous logging of throughput, face reflection and gas composition; and (iv) a low-cost, deployable enclosure qualified against the same measurements.

4 Aim, research questions and hypotheses

4.1 Aim

The aim of the project is to **determine whether, and under which atmospheric conditions, the UV-induced degradation of single-mode fibre end faces carrying a few milliwatts of 369.5 nm light can be stopped, and to demonstrate a low-cost, 3D-printed end-face enclosure that achieves this in routine use**. The working hypothesis of the group is that removing oxygen from the end faces halts the degradation. The project is designed so that it gives a clear answer whether this hypothesis is right or wrong. If oxygen turns out not to be the culprit, the same measurements identify what is (Table 4).

4.2 Research questions

RQ1 Where and what? Is the throughput loss located at the input face, the output face or in the fibre, and what is its physical nature: a deposit, a restructured silica surface, colour centres, or misalignment?

RQ2 Is oxygen the cause? At fixed (low) humidity and contaminant levels, does removing O_2 change the degradation rate, by how much and in which direction, and is there a threshold O_2 concentration?

RQ3 What else matters? What are the separate roles of water vapour, organic and siloxane vapours (including those emitted by 3D-printed parts, adhesives and cable jackets), airborne particles and, optionally, hydrogen?

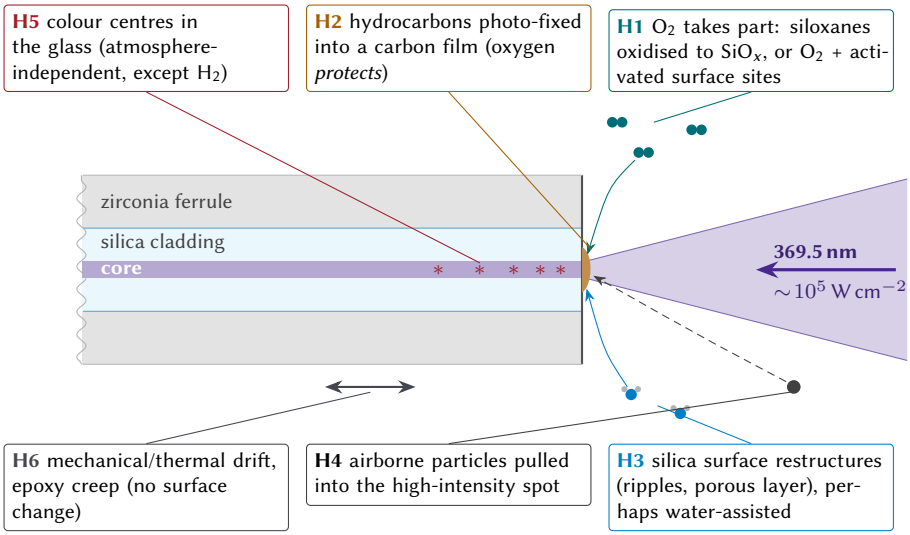


Figure 1: Candidate processes at a UV-illuminated fibre end face (input end shown; the output face sees practically the same guided intensity). Only H1–H4 can depend on the surrounding gas. H5 can be influenced only by hydrogen, which diffuses into silica; H6 is an artefact. The experiment is designed to discriminate between them (Table 4).

RQ4 Does it work in practice? Can a passive, sealed 3D-printed cell hold the required atmosphere for months, and does it stop the degradation of a working 369.5 nm fibre link in one of the group's experiments?

4.3 Competing hypotheses

The observations available so far constrain the mechanism considerably: high throughput after re-polishing, a decline whose rate increases with power, and a decline at *both* ends. Re-polishing removes only micrometres to tens of micrometres of glass but fully restores the throughput, so the damage sits at, or within micrometres of, the end faces. Bulk colour-centre formation along the fibre would not be undone by polishing. The guided intensity is nearly the same at both faces, so a process driven by the local optical intensity at a glass surface explains why the output face degrades too. Mechanical misalignment of the input coupling does not. The literature reviewed in Section 3 adds a candidate the group may not have considered: the silica surface itself can restructure under CW blue light [14]. Six hypotheses are retained (Figure 1):

H1 – oxygen-assisted deposition or surface chemistry
Oxygen takes part in the damaging reaction. The best-documented route is the photo-oxidation of siloxanes, released by silicone greases, rubbers and adhesives, into non-volatile SiO_x at the high-intensity spot. In a fibre-coupled 410 nm module this was *accelerated* above 100 ppm O_2 [89]. A second, more speculative route is the reaction of O_2 with photo-activated sites on the polished silica. *Prediction:* removing O_2 slows or stops the degradation, and the rate rises with the O_2 fraction. Any deposit is rich in Si and O and resists solvents, and a silicone source in the cell accelerates degradation far more in clean air than in nitrogen.

H2 – carbonaceous laser-induced contamination
Hydrocarbons and plasticisers adsorbed from the gas phase are photo-fixed into an absorbing, carbon-rich film. In every community that has studied this, oxygen, and often water vapour, suppresses or removes such deposits [55, 71, 73]. Growth can be fastest at *trace* oxygen levels [74]. *Prediction:* clean gas slows the degradation, but removing O_2 while organics are present (e.g. outgassing from printed parts) makes it *worse*, especially at 1 – 1000 ppm residual O_2. Humid gas may help.

Table 4: Predicted degradation rate in each test atmosphere relative to the ambient-air control, for each hypothesis. **AIR**: ambient lab air; **CDA**: clean, dry synthetic air; **N2**: clean, dry nitrogen; **ABS**: sealed printed cell with O_2 absorber; **WET-N2**: humidified N_2; **FG**: forming gas (5 % H_2 in N_2); **N2+P**: nitrogen with a coupon of printing polymer in the cell; **+SIL**: a silicone coupon in both a **CDA** and an **N2** cell. Symbols: ● similar to **AIR**; ○ strongly reduced; ↑ faster than **AIR**; ●/○ depends on a secondary factor (residual organics for H2, water for H3); – no clear prediction. **CDA** versus **N2** is the decisive test of the oxygen hypothesis. The deposit chemistry and the **+SIL** pair tell H1 from H2.

	AIR	CDA	N2	ABS	WET-N2	FG	N2+P	+SIL	extra signature
H1 O_2-assisted	●	●	○	○	○	○	○	CDA ≫ N2	rate rises with O_2; Si/O-rich deposit
H2 carbon LIC	●	○	●/○	●/↑	○?	–	↑	N2 ≥ CDA	C-rich, fluorescent film; fastest at trace O_2
H3 silica restructuring	●	●/○	●/○	●/○	●	–	●	=	ripples along polarisation; only Si and O
H4 particles	●	○	○	○	○	○	○	=	discrete particles; partly wiped off
H5 colour centres	●	●	●	●	●	○?	●	=	no surface change; defect PL bands
H6 mechanical/thermal	●	●	●	●	●	●	●	=	recovered by re-peaking

H3 – restructuring of the silica surface
The glass surface itself develops nanostructures, for example laser-induced periodic surface structures (LIPSS) and a porous, lower-density silica layer. These have been found on both faces of single-mode fibres carrying CW 405 nm light, with no foreign elements present [14, 49]. Adsorbed water or surface charging may assist the process [49]. *Prediction:* weak atmosphere dependence (perhaps slower in dry gas), a pronounced intensity threshold, and ripples aligned with the polarisation visible in SEM or AFM.

H4 – particle accumulation
Airborne particles and oil droplets are pulled into the high-intensity spot by optical gradient forces and then burned into the surface [46]. *Prediction:* any filtered gas helps, and discrete particles are visible on the end face.

H5 – colour centres in the glass
Point defects form in the core within the polished-away depth, for example in a defect-rich layer left by polishing. *Prediction:* no atmosphere effect, except a possible reduction in hydrogen-containing gas [31].

H6 – mechanical or thermal artefact
Drift of the coupling optics, epoxy creep or ferrule movement. *Prediction:* no atmosphere effect, and throughput is recovered by re-peaking the alignment without touching the fibre.

Several mechanisms may act together. H1 and H2 can even coexist, with oxygen helping against one contaminant and harming with another. H2 and H4 are hard to separate in lab air, and any absorbing film also heats the end face (Appendix A), which can turn a slow process into a runaway one. The experiment is therefore designed to measure the effect of each atmospheric factor separately, not just to compare "with" and "without" oxygen.

Why oxygen can only act at the surface

At room temperature molecular oxygen is practically immobile in silica glass. With literature diffusion data it penetrates no more than nanometres even in years (Appendix A), so the external atmosphere cannot change bulk solarisation of the core. Hydrogen is different: it diffuses through a 125 µm fibre within days [37]. Any oxygen effect must therefore act at the end faces. The observed behaviour

of the group's fibres (restored by re-polishing, both ends affected) points to exactly such surface processes, which makes the project well motivated.

5 Methodology and experimental design

5.1 Design principles

Fibre degradation experiments are slow, and many things can change in a lab over several weeks. The design therefore follows five rules:

P1 **Parallel, not sequential.** All test atmospheres run *at the same time* from one light source, in identical optical arms. Laser drifts, lab humidity and seasonal effects then affect all arms equally and cancel in the comparison.

P2 **Change one factor at a time.** The decisive pair of arms differs *only* in the presence of O_2: clean dry synthetic air versus clean dry N_2, both passed through the same purifier and particle filter (Table 5).

P3 **Separate the surface from the alignment.** Throughput is recorded before and after re-peaking the input coupling, and the back-reflection from the input face is logged continuously. Alignment drift (H6) can then be told apart from changes of the end face.

P4 **Locate the loss.** At the end of every run a fixed sequence of non-destructive and destructive steps (Section 5.7.1) splits the loss between input face, output face and fibre.

P5 **Decide in advance.** Metrics, the analysis and the decision thresholds (Section 5.9) are written down before the confirmatory campaign starts. This guards against reading patterns into noisy data after the fact.

5.2 Phase 0: forensics of fibres already degraded in the lab

The cheapest information is already in the lab. In the first two months the student collects degraded fibres from the group's experiments (ideally 3–6, with as much history as possible: power, hours of use, wavelength, environment, cleaning and polishing history, PC or APC ends) and:

1. images both end faces *before any cleaning* with the fibre inspection microscope (core zone, standard magnification and illumination, Appendix B);
2. measures throughput at low power ($\leq$0.3 mW) to avoid changing the surfaces further;
3. runs the cleaning sequence of Section 5.7.1 on half of the fibres and keeps the others uncleaned for surface analysis (optical and fluorescence microscopy, SEM/EDX, AFM or white-light interferometry, Raman, and where available ToF-SIMS or XPS).

In parallel, the student surveys likely contaminant sources near the group's fibre ends: silicone grease, silicone tubing and adhesives, thermal paste, cable jackets, printed parts and cleaning practices. Within weeks this shows whether the damage is a *deposit* (and whether it is rich in carbon or in silicon), a *surface modification* or *nothing visible*. That result already favours some hypotheses over others and decides which test atmospheres are most informative.

5.3 Test fibres and end-face preparation

- **Fibre type.** The same fibre as used in the experiments, so that the results transfer directly: Thorlabs P3-375PM-FC patch cables, FC/APC at both ends, built on PM-S350-HP pure-silica-core PANDA fibre (Section 3.3), ideally from a single batch. Thorlabs sells the bare fibre no longer, but a fibre with the same part number and specification is available from Coherent (Nufern). Cheaper non-PM SM300 cables, which also have a pure-silica core, can serve as extra arms where polarisation does not matter.

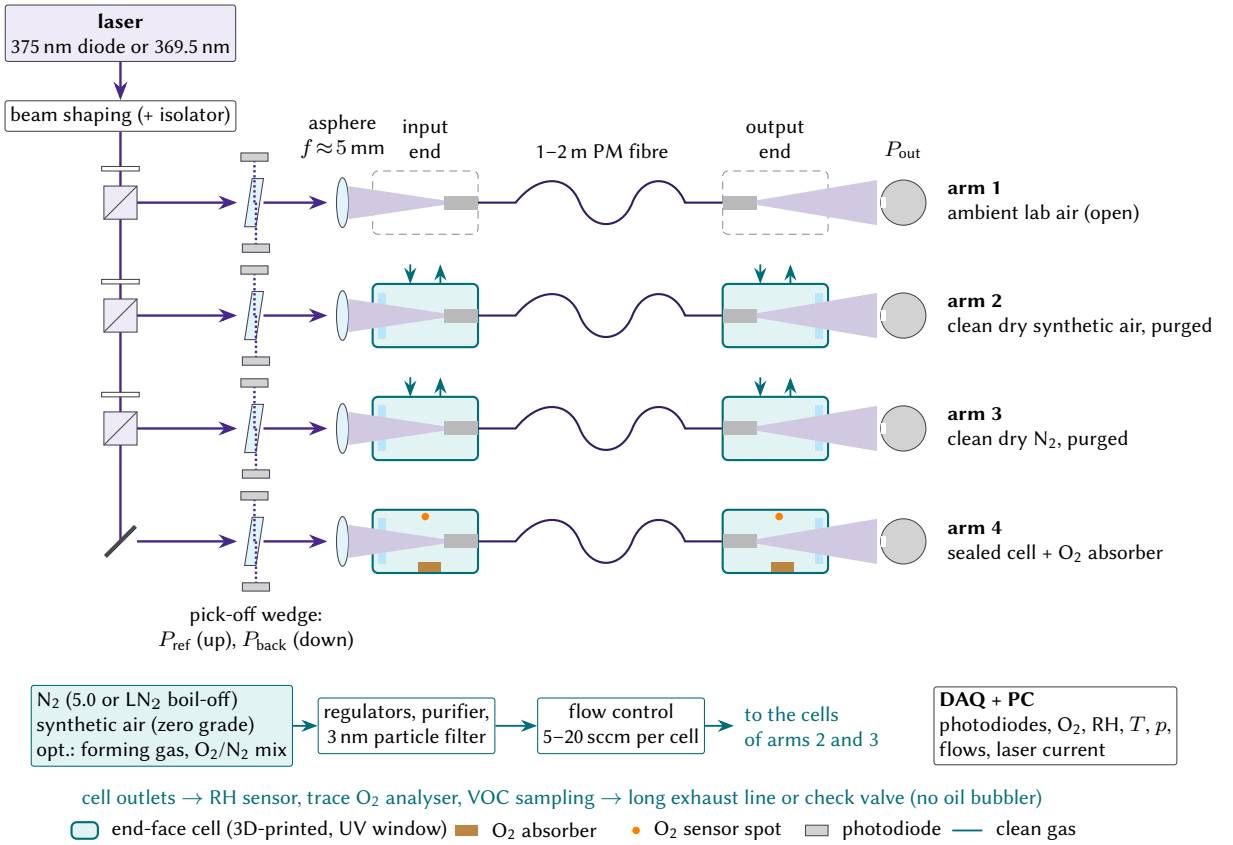


Figure 2: Multi-arm accelerated-ageing testbed. One near-UV source feeds identical arms that differ only in the atmosphere at the two end faces. Each arm records the reference, back-reflected and transmitted powers. Purged arms share one gas-purification chain; the static arm uses an oxygen absorber and an optical O_2 sensor spot. Arm 1 has no cells and reproduces the present lab conditions.

- **Format.** 1 – 2 m lengths, connectorised and polished in-house with one documented procedure and one epoxy (or bought as identical patch cables). The fibre is coiled loosely (bend radius well above the specified minimum) and taped to the table, so that bend loss cannot change.
- **Acceptance.** Every end face is inspected and imaged after polishing, and its geometry is measured if an end-face interferometer is available. Its initial coupling efficiency is measured at low power. Ends that miss the acceptance criteria (no defects in the core zone; coupling within a few per cent of the batch median) are re-polished before use.
- **Surface preparation is a variable.** At 405 nm, cleaved faces degraded four times more slowly than glued-and-polished ones [49]. Every test end is therefore prepared by exactly the same polishing procedure. A comparison of preparation methods (e.g. an additional final polishing step) is an optional arm in Campaign II.
- **Traceability.** Each fibre gets an ID. Each end has its own polishing record and image set. The assignment of fibres to arms is randomised, and arm positions are swapped between runs so that any arm-specific optical problem cannot look like an atmosphere effect.

5.4 Light source and multi-arm testbed

Figure 2 shows the testbed. Its main elements are:

Source. A dedicated 375 nm single-mode laser diode (Section 3.6) with anamorphic beam shaping. It is protected from feedback by an optical isolator if one is available at this wavelength, otherwise by tilted optics; the APC faces do not reflect back into the beam anyway. The dedicated source frees the group's 369.5 nm lasers for experiments and allows long, uninterrupted runs. The photon energies at

375 nm and 369.5 nm differ by only 1.5 %, and both lie on the same flanks of the weak defect bands of Table 1. Equivalent behaviour is therefore expected, but this assumption is *tested* in Phase 4, which runs at 369.5 nm. If no diode is available, a pick-off of a few tens of milliwatts from an existing 369.5 nm system can feed the testbed.

Power distribution. A cascade of half-wave plates and polarising beam splitters divides the beam into 4–6 arms with adjustable and equal powers. A further half-wave plate in each arm aligns the polarisation to the slow axis of the PM fibre. The operational test power is 3 mW guided; the accelerated condition is 10 mW. A sentinel arm at $\leq$0.1 mW in N_2 measures the drift of the measurement system itself.

Coupling. An aspheric lens with near-UV anti-reflection coating ($f \approx 4.5 - - 8\,\mathrm{mm}$, chosen for mode matching, Appendix A) focuses the beam through the cell window onto the fibre end face, at the 3.8° incidence an 8° APC face requires. Commercial launch systems for FC/APC (e.g. Thorlabs FiberPort PAF2A-5A or -7A) or a moulded asphere (e.g. C230TMD-A) can be used. The high-index glasses of such aspheres transmit less well in the near UV, so their transmission at 369.5 nm should be checked. The lens, or the cell, sits on a stable XYZ mount. Automated peak-up with motorised mirror mounts is a stretch goal; the baseline is a manual re-peak once a day with a fixed procedure.

Detection. A wedged pick-off in each arm feeds a reference photodiode (P_{ref}) and a second photodiode that records the light reflected by the input face (P_{back}; with FC/APC connectors this detector sits in the specularly reflected beam, which does not retrace the input). The output power P_{out} is measured with an integrating-sphere or diffuser-fronted photodiode, which is insensitive to small changes of the output beam. Silicon photodiodes lose UV responsivity and gain dark current under prolonged UV exposure [137, 142]. All detectors are therefore operated at low intensity (attenuated, diffused or shuttered), photodiodes sold for UV resistance are preferred (e.g. Hamamatsu S12698 series [139]), and a reference power meter that is otherwise kept in a drawer cross-checks them weekly. SiC photodiodes, the usual choice for UV-stable detection, are not suitable here: 369.5 nm lies at the edge of their spectral response.

Logging. All photodiodes, laser current and temperature, lab and arm temperatures, humidity, pressure, flows and O_2 readings are logged at 1 Hz (stored as 1-minute means) by a Python data logger (Appendix C). Every manual intervention is recorded in an event log with a timestamp.

5.5 Test atmospheres

Table 5 lists the atmospheres. Purged arms receive 5 – 20 sccm per cell through a common chain of regulator, oxygen and moisture purifier and 3 nm particle filter. This is far less than the 0.5 – 1 L/min that Marciniak et al. blew through open collimators to keep particles out [46], but a sealed cell needs no more than a small positive outflow (Appendix A). Gas lines are stainless steel or copper, with only short pieces of nylon. One metre of polyurethane or fluoropolymer tubing lets in tens of ppm of O_2 at these flows (Appendix A). Each outlet passes a humidity sensor before a long, thin exhaust line or a check valve that stops back-diffusion. Oil bubblers are avoided, because oil vapour is itself a contaminant. A trace O_2 analyser is switched between the outlets of the purged arms. Low-cost metal-oxide VOC sensors need oxygen to work, so they are used only on oxygen-containing outlets. The oxygen-free streams are sampled onto sorbent tubes for TD-GC/MS analysis. The synthetic air is ordered with a specified hydrocarbon content of $\leq 0.1\,\mathrm{ppm}$, because "zero grade" alone can mean 5 ppm [115]. O_2 mixtures for **O2-x** are made by adding synthetic air to N_2 with two flow controllers, or are bought as certified mixtures. Humidified arms use a water bubbler in a split stream. The terms "getter" and "absorber" are used carefully. Vacuum getters are unsuitable inside a cell at atmospheric pressure; the cells use absorbers of the kind used in food and museum conservation (Section 3.5).

5.6 End-face cells

Each fibre end that is not in ambient air sits in a small cell (Figure 3). The requirements are:

Table 5: Test atmospheres. Core arms run in Campaign I; extended arms are selected for Campaign II on the basis of Campaign I and the Phase 0 forensics.

Arm	Atmosphere at both end faces	Purpose
Core arms		
AIR	ambient lab air, ends open as in the experiments	control; reproduces the problem
CDA	synthetic air (20.9 % O_2 in N_2, zero grade), purified, filtered, dry	contains O_2 but no contaminants, particles or water
N2	N_2 (grade 5.0 or LN_2 boil-off), same purifier and filter, dry	identical to **CDA** except for O_2 ⇒ decisive test of H1 vs H2
ABS	sealed printed cell, pre-purged with N_2, O_2 absorber that also takes up water (RP-A type), no flow	the low-cost, passive solution to be deployed
Extended arms		
O2-x	N_2 with $x = 10$, 100, 1000, 10 000 ppm O_2	dose–response, sets the O_2 specification for the cells
WET-N2	N_2 humidified to 30 – 50 % RH	water without O_2: assists damage (H3) or protects against carbon deposits (H2) [73, 74]
WET-CDA	synthetic air at lab-like RH, filtered	lab air minus contaminants and particles (H2, H4)
FG	forming gas, 5 % H_2 in N_2	hydrogen passivation of defects (H5)
N2+P	N_2 cell containing a coupon of the printing material	are printed parts a contamination source? (H2)
+SIL	a silicone coupon in one **CDA** and one **N2** cell	oxygen-assisted siloxane deposition (H1) [89]
VAC	vacuum (optional, if a chamber with fibre feedthrough is free)	link to in-vacuum fibre ends

1. **Optical access** through a UV fused-silica window, AR-coated near 370 nm and slightly wedged and tilted to avoid etalons and back-reflections into the fibre. The window lies in the converging beam well away from the focus, so the intensity on it is low. For a 1 mm window at NA 0.1 the added spherical aberration is negligible (Appendix A).
2. **A small, swept volume** (10 – 30 cm^3). Gas enters next to the end face and leaves downstream, so that anything released by the connector, epoxy or cable is carried *away* from the face.
3. **Tightness.** The whole connector sits inside the cell and the cable leaves through an epoxy-potted gland. Otherwise air diffuses along the jacket to the ferrule. Each cell must pass a pressure-decay leak test (target $< 1 \times 10^{-5}$ mbar L/s, Appendix B); at that leak rate a typical absorber sachet lasts for months (Appendix A).
4. **Static variant (ABS).** The absorber sits in a pocket behind a porous PTFE membrane, out of line of sight of the end face, so that no dust or vapour from the sachet reaches it. The cell is first purged with N_2 to about 1 % O_2. The absorber then only has to remove a small residue, and the pressure in the sealed cell drops by about 10 mbar instead of 210 mbar, which would otherwise draw air in through any leak. For a dry cell the absorber is of the type that also takes up water (RP-A). Self-reacting iron types humidify the cell and are reserved for humid variants, or are combined with a molecular-sieve desiccant [101, 103].
5. **Water budget.** Printed walls let in far more water than oxygen. Through 60 cm^2 of 3 mm PLA or ABS, about 4 mg of water per day reaches a cell that holds only 0.6 mg at saturation (Appendix A).

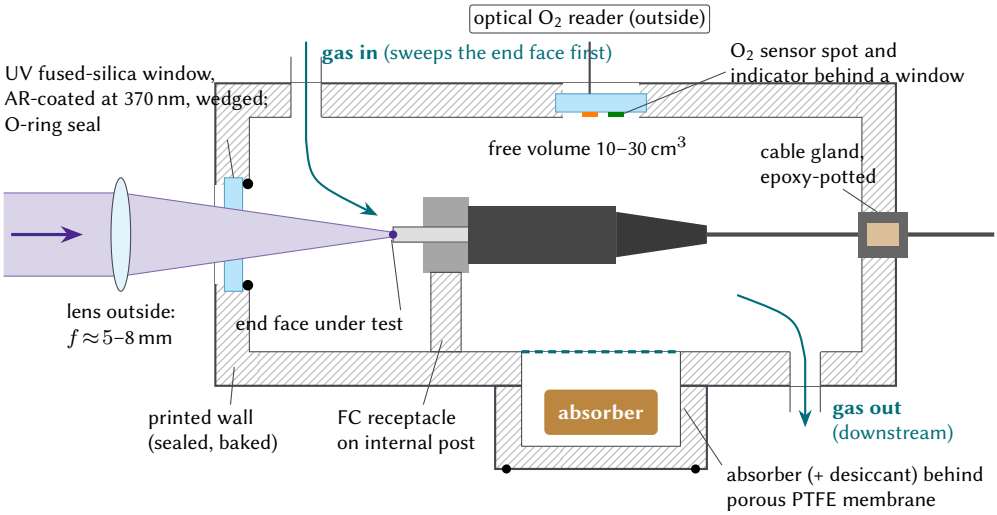


Figure 3: Concept of a 3D-printed end-face cell (cross-section, input end). The whole connector is enclosed and the cable is sealed in a potted gland, because air would otherwise diffuse along the cable to the ferrule. Gas enters next to the end face and leaves downstream. In the static variant the inlet and outlet are closed after an N_2 pre-purge and the absorber in the pocket removes the residual oxygen. The window sits in the converging beam, far enough from the focus that the intensity on it is low.

Static cells therefore carry a desiccant with months of capacity, the walls are made of a low-permeability material, and RH is logged.

6. **Verification.** An ultra-trace optical O_2 sensor spot (detection limit ∼0.5 ppm, read through a window so that no electronics sit inside the cell and shielded from the UV beam) and a colour indicator for gross failures monitor the atmosphere. Each cell is qualified by its rate of rise. With an FKM window O-ring alone, O_2 would rise by tens of ppm per day in a sealed 30 cm^3 cell without an absorber (Appendix A).
7. **Clean materials.** Printed parts are a possible source of organic vapour (H2). Prototypes are printed in PETG (low outgassing, low VOC emission) with 100 % infill, walls of at least 1.1 mm, slight over-extrusion and randomised seams [117]. They are sealed with a low-outgassing epoxy and baked under N_2 purge. Final cells use a low-outgassing polymer (PEEK, ULTEM or Antero by FDM; Rigid 10K by SLA) or machined aluminium (Table 3). ABS and standard SLA resins are avoided. The **N2+P** arm tests directly whether the chosen material harms the end face. O-rings are FKM or butyl, seated on machined, ironed or SLA-printed lands, and fitted dry or with a minimal film of PFPE grease. Silicone O-rings, silicone grease and silicone adhesives are banned, because they release siloxanes (H1). The gland is potted with a low-outgassing epoxy.

The student designs the cells in CAD with the dimensions of the group's fibre mounts. Two designs are worth prototyping early: (a) a cap that encloses the connector and screws onto a standard fibre-launch receptacle, and (b) a box that encloses the whole launch (lens included). Design (a) has a smaller volume; design (b) is easier to seal.

5.7 Diagnostics

Table 6 summarises the measurements. Two of them are cheap but very useful. First, the *power reflected by the input face* is 3.7 % of the incident power. With the 8° FC/APC faces of the group's cables this reflection does not retrace the input beam. It leaves ≈24° away from it (Appendix A) and can be caught with a small mirror or photodiode. Even a thin film changes it through thin-film interference and scattering, so it is a sensitive in-situ monitor of the input face. Second, *organic films fluoresce* under near-UV excitation, while clean silica hardly does. A long-pass filter and a small spectrometer that looks back through the coupling lens can therefore see a film start to grow long before the throughput drops noticeably.

Table 6: Measurements, their purpose and cadence.

Measurement	What it tells us	When
$\eta = P_{\text{out}}/P_{\text{ref}}$ (normalised)	total throughput change of each arm	continuous, 1 Hz
re-peak gain $\eta_{\text{after}}/\eta_{\text{before}}$	alignment drift (H6) vs end-face change	daily or automated
$P_{\text{back}}/P_{\text{ref}}$	optical change of the input face (film, roughness)	continuous
O_2, RH, VOC index, T, p, flow	that each atmosphere is what it should be	continuous
output beam profile and power through a mode-selective aperture	loss of beam quality (rises in M^2 have been seen at 405 nm [49]) that a large detector does not register	every 30 min
visible luminescence from the input face (spectrometer, optional)	fluorescence of growing organic films; defect PL bands	every 30 min
end-face microscopy (bright field, fluorescence)	deposits, particles, damage	start and end of each run
cleaning sequence (Section 5.7.1)	splits the loss between faces and bulk	end of each run
SEM/EDX, AFM or interferometry, Raman, ToF-SIMS/XPS	composition and thickness of films	selected ends

5.7.1 End-of-run cleaning sequence

At the end of each run the transmission is measured at low power after each of the following steps, always with a re-peak and first at the output end, then at the input end:

1. no-touch inspection and imaging (preserves evidence);
2. dry clean (filtered gas jet, then a dry cleaning tool);
3. solvent clean (reagent-grade isopropanol, lint-free wipe; acetone as a second solvent);
4. oxidative clean (O_2 plasma or UV/ozone, if available), which removes organic but not inorganic (e.g. siloxane-derived SiO_x) films;
5. re-polish, recording the removed length.

The loss recovered at each step assigns the damage to a face and to a category (loose particles, soluble film, oxidisable film, bonded or structural damage). Any loss left after both ends have been re-polished is attributed to the fibre itself (H5). Where an arm has two fibres, one goes through this sequence and the other is kept uncleaned for surface analysis.

5.8 Experimental campaigns

Phase 1 – baseline in ambient air (months 4–5)
Three to four **AIR** arms at 1 mW, 3 mW and 10 mW, plus the sentinel. Outputs: (i) proof that the testbed reproduces the degradation seen in the lab; (ii) the power law $r \propto P^n$ of the initial degradation rate. An exponent $n \approx 2$ points to a two-photon process; (iii) fibre-to-fibre scatter; (iv) the run length needed in the later campaigns. Raising the power only accelerates the test if the mechanism stays the same. Surface restructuring at 405 nm was reported above $\sim$0.5 MW/cm^2 [14], close to the 0.44 MW/cm^2 reached here at 10 mW. The end faces from the 3 mW and 10 mW arms are therefore compared by microscopy before 10 mW is used as an accelerated condition.

Phase 2 – Campaign I, atmosphere screening (months 6–7)
AIR, **CDA**, **N2** and **ABS** in parallel at the accelerated power (or 3 mW if Phase 1 shows that 10 mW

changes the mechanism, e.g. thermal runaway). Each run lasts until **AIR** has lost ∼30 % or for three weeks, whichever comes first. If time allows, a second run with permuted arm positions follows.

Phase 3 – Campaign II, confirmation and mechanism (months 8–9)
Repeat the key contrast at 3 mW with at least two fibres per condition, and add the most informative extended arms (Table 5): the O_2 dose–response if O_2 matters, **WET-N2**/**WET-CDA** and **N2+P** if it does not, and the **+SIL** pair if the forensics find silicon-rich deposits. A crossed configuration (input in N_2, output in air, and vice versa) tests whether both faces respond in the same way. Marciniak et al. observed that the purged and unpurged ends of the same fibre behaved differently [46].

Phase 4 – demonstration at 369.5 nm (months 9–11)
The best cell design is fitted to both ends of a working 369.5 nm fibre in one of the group's experiments, with a control fibre or the fibre's own history as reference, for at least eight weeks. This phase also checks the transfer from 375 nm to 369.5 nm and whether the cell is practical in use (alignment stability, access, window losses).

How the outcome of Campaign I steers the project

- **N2 and ABS stable, CDA degrades like AIR**: oxygen takes part (H1). Check the deposit chemistry: if it is rich in Si and O, siloxanes are involved, and their sources near the fibres should be removed anyway. Then measure the O_2 dose–response, specify the cell atmosphere and deploy **ABS** cells. This is the outcome the group hopes for.
- **CDA and N2 both stable, AIR degrades**: something in lab air other than O_2 is responsible: particles, organic vapours or water (H4, H2, or a water-assisted H3). Campaign II separates them. A sealed or purged *clean* cell solves the practical problem, with or without oxygen. This is what the published nitrogen-purge results would suggest [46].
- **CDA stable, N2 or ABS degrade**: carbonaceous contamination with oxygen as a protective agent (H2). This is the outcome that space-laser, UV-cavity and laser-diode experience predicts (Section 3.4). Oxygen absorbers would be counter-productive; the recommendation becomes a clean, dry (or humidified) air purge.
- **No atmosphere helps**: restructuring of the silica, colour centres or an artefact (H3, H5, H6). Shift the effort to end-face and fibre engineering. Coreless end caps lower the intensity at the glass–gas interface by two to three orders of magnitude (Appendix A). Other options are hydrogen-loaded and UV-cured fibre and hollow-core fibre (Section 3.3).

Every branch gives a publishable answer and a concrete recommendation for the group.

5.9 Data analysis and decision rules

Metrics. The primary quantity is the normalised, re-peaked throughput

$$\eta(t) = \frac{[P_{\text{out}}/P_{\text{ref}}](t)}{[P_{\text{out}}/P_{\text{ref}}](t_0)},$$

corrected for detector temperature coefficients, with flagged intervals (interventions, outages) excluded. The *primary endpoint* is the initial degradation rate $r = -\mathrm{d}\eta/\mathrm{d}t$ in % per 1000 h at 3 mW. Secondary endpoints are the time to 10 % loss, the fit parameters of a phenomenological model (e.g. a stretched exponential $\eta = 1 - A\{1 - \exp[-(t/\tau)^{\beta}]\}$ in time or dose), the change of P_{back}, and the outcome of the cleaning sequence.

Statistics. Confidence intervals come from a block bootstrap, which allows for autocorrelated data. Conditions are compared through rate ratios. With replicate fibres, a mixed-effects model with fibre as random effect separates the atmosphere effect from fibre-to-fibre scatter. Phase 1 provides the scatter needed to choose the number of replicates.

Decision rules (fixed before Campaign II).

- **O_2 effect established:** r_{N2}/r_{CDA} differs from 1 by at least a factor of two, with the 95 % confidence interval excluding 1, in two independent runs.
- **Degradation halted:** the 95 % upper limit on $|r|$ is below 1 % per 1000 h at 3 mW (equivalent to $< 10\,\%$ loss per year of continuous use), at least twenty times below **AIR**, with no visible change of either end face. With a daily scatter of 0.3 % this limit can be reached in three to six weeks (Appendix A).

5.10 Success criteria and deliverables

S1 The location and physical nature of the degradation are established for the group's fibres (RQ1).

S2 The effect of O_2 on the degradation rate is measured with a stated confidence interval, or excluded at the level above (RQ2), and the roles of water, contaminants and particles are tested (RQ3).

S3 A cell design is qualified: $O_2 < 100\,\mathrm{ppm}$ (target $< 10\,\mathrm{ppm}$) and RH $< 5\,\%$ for at least eight weeks without maintenance. It is tested on a working 369.5 nm link (RQ4). If the atmosphere helps, the degradation-halted criterion is met.

S4 Open deliverables: CAD files and bill of materials of the cells, standard operating procedures, the data set with analysis code, the MSc thesis, and (if the results allow) a short paper.

6 Work plan

The plan assumes a full-time, 12-month project (Figure 4). Items that need ordering are listed in Section 7 and should be ordered in month 1, because optics, gas fittings and sensors can have lead times of several weeks. A compressed 6-month version is given at the end of this section.

6.1 Decision points

D1, end of WP4: is the degradation fast enough to study?
If **AIR** at 3 mW loses less than about 5 % in two weeks, the campaigns run at 10 mW. If necessary the intensity is raised further with a smaller mode-field fibre or by running two arms in series on one source. If the testbed does not reproduce the lab degradation at all, the difference between testbed and lab (wavelength, contaminant sources near the experiments, cleaning practice) becomes a research question in itself, and the forensic results guide the next steps.

D2, end of WP5: which mechanism?
Follow the branches of the decision box in Section 5.8.

D3, end of WP6: what is deployed?
The cell type (purged or static, with or without oxygen) for the Phase 4 demonstration follows from the confirmatory result.

6.2 Compressed 6-month variant

If the project has to fit into six months, keep the logic but drop breadth: month 1: inductions, forensics, ordering; month 2: two to four arms and simple purged cells (no static cell yet); month 3: short baseline at 10 mW; months 4–5: one campaign with **AIR**, **CDA**, **N2** (plus **ABS** if ready); month 6: cleaning sequence, analysis, writing. The demonstration on a working link and the extended arms become recommendations for a follow-up project.

Minimum viable thesis

Even if the full testbed is delayed, the following is enough for a good thesis: the forensic analysis of used fibres, plus one well-controlled parallel comparison of ambient air, clean dry air and dry N_2 at

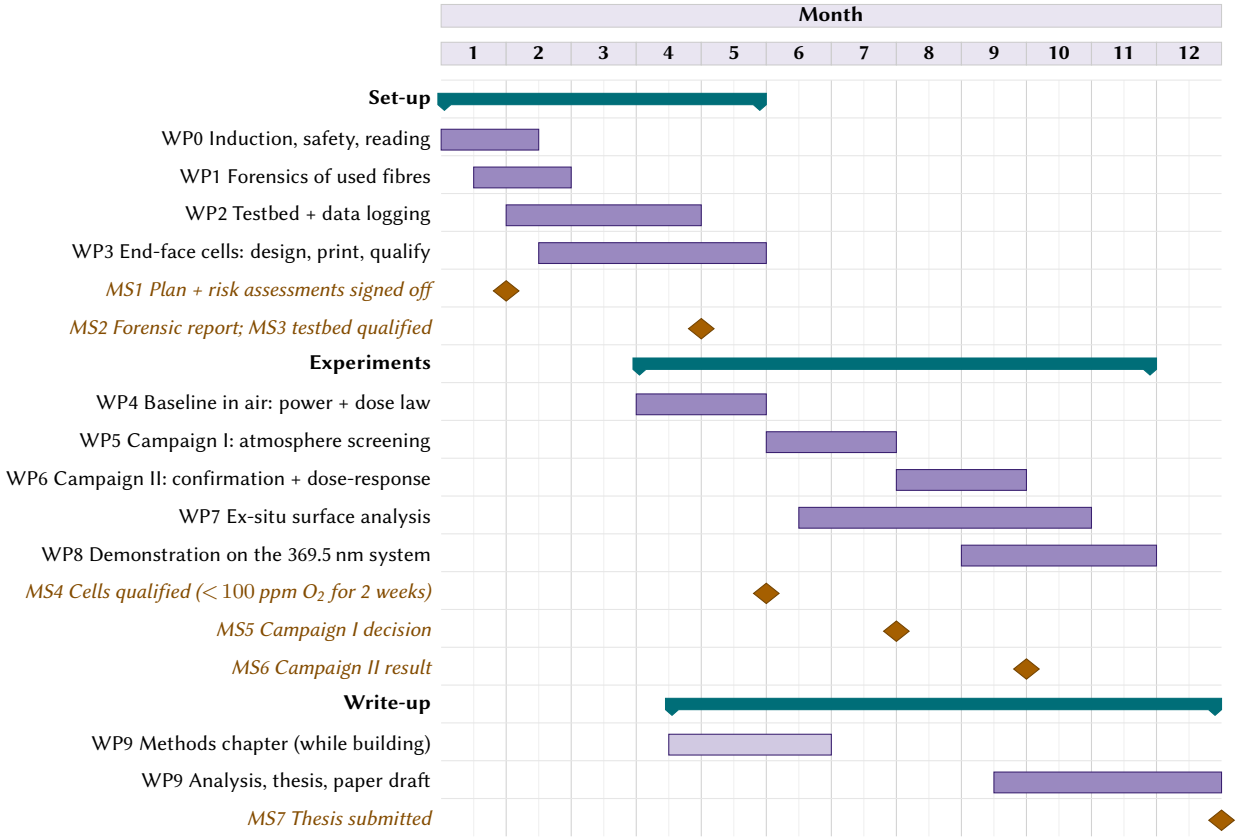


Figure 4: Work packages (WP) and milestones (MS) for a 12-month project in half-month resolution. The methods chapter is drafted while the testbed is being built; the analysis and thesis writing overlap with the demonstration phase.

the end faces, with the cleaning sequence at the end. The extended arms, the static absorber cells and the demonstration are the parts that turn it into an excellent thesis and a paper.

Table 7: Work packages, main outputs and milestones.

WP	Content	Output / milestone
WP0	inductions (laser, gas, chemical, workshop), literature, project plan	MS1: plan, risk assessments and order list agreed (end of month 1)
WP1	forensics of degraded fibres from the lab (Section 5.2)	MS2: short forensic report with images; choice of extended arms
WP2	testbed: source, splitting, 4–6 arms, detectors, logger, sentinel arm	MS3: testbed qualified (drift of sentinel $< 0.3\,\%$/day; logging robust for one week)
WP3	end-face cells: CAD, printing, sealing, leak and O_2 hold tests, material screening	MS4: cells qualified (< 100 ppm O_2 for two weeks, leak rate $< 1 \times 10^{-5}$ mbar L/s)
WP4	Phase 1 baseline in air: power and dose dependence, fibre scatter	degradation law; run length for later campaigns
WP5	Phase 2 / Campaign I: **AIR**, **CDA**, **N2**, **ABS**	MS5: decision on Campaign II (decision box in Section 5.8)
WP6	Phase 3 / Campaign II: replicates, extended arms, crossed input/output	MS6: confirmatory result against pre-registered criteria
WP7	ex-situ surface analysis of selected ends	identification of films or damage
WP8	Phase 4: cells on a working 369.5 nm link for ≥ 8 weeks	demonstration result, practical guidelines
WP9	methods chapter (months 4–6), analysis, thesis, paper draft	MS7: thesis submitted

7 Resources and indicative budget

7.1 What the group is likely to have already

Check these in month 1: fibre polishing and inspection equipment; spare 369.5 nm power on one of the laser systems (for Phase 4 and as a fall-back source); optical mounts, mirrors, wave plates and beam splitters for the near UV; power meters; the laboratory N_2 supply; 3D printers (FDM, possibly SLA) in the group or the workshop; a fusion splicer (for end caps, optional); and access to SEM/EDX, AFM or white-light interferometry, Raman and XPS/ToF-SIMS through university facilities.

7.2 Items to buy

Table 8 lists the main items. Thorlabs prices are US list prices read on 18 October 2026 (UK prices and university discounts differ). Items marked ≈ are rough estimates that should be replaced by quotes.

Table 8: Main items for the testbed and cells (US dollars).

Item	Example and quantity	Cost
Light source and distribution		
375 nm single-mode diode	Thorlabs L375P70MLD (Nichia NDU4116, 70 mW) [136]	5 589
TEC mount	Thorlabs LDM56F (supports the F pin code)	810
diode current and TEC controller	one combined unit	≈ 2 000–3 000
splitting and steering optics	half-wave plates, PBS cubes, mirrors, mounts for 4–6 arms	≈ 2 000–4 000

Item	Example and quantity	Cost
fibre launches	4–6 × FiberPort PAF2A-5A or -7A (FC/APC) at $674	2 700–4 000
Test fibres and preparation		
test cables	8–12 × P3-375PM-FC-2 at $366 (plus spares)	2 900–4 400
optional control fibres	4 × P3-305A-FC-1 (SM300, FC/APC) at $131	530
polishing consumables	diamond films 30–1 µm, 0.01 µm final film, FC/APC puck	≈ 500
Detection and logging		
reference power meter	S120VC sensor + PM100USB	1 061
arm photodiodes and DAQ	12–18 UV-resistant photodiodes, transimpedance amplifiers, multichannel ADC	≈ 1 500–3 000
optional integrating sphere	S142C for the output detector of a key arm	1 215
End-face cells		
windows	8–12 × WG41010-A (UV fused silica, 1 mm, AR 350 – 700 nm) at $123	990–1 480
FC/APC receptacles, sleeves	ADAFCPM2 ($62) or equivalent, 4–8	250–500
printing materials, O-rings, epoxy, inserts, PTFE membranes, desiccant	PETG, FKM/butyl O-rings, low-outgassing epoxy	≈ 500–1 000
Atmosphere control and verification		
gas handling	regulators, O_2/H_2O purifier, 3 nm filters, flow meters or mass-flow controllers, stainless lines and fittings	≈ 3 000–6 000
trace O_2 sensing	reader with ultra-trace sensor spots (e.g. PreSens SP-PSt9 [107]); optionally a galvanic trace analyser for exhausts	≈ 3 000–8 000
absorbers and indicators	RP-A and AGELESS sachets, indicators	≈ 200
gases	synthetic air (≤ 0.1 ppm hydrocarbons), small N_2 6.0 cylinder, optional mixtures, rental	≈ 500–1 500/yr
Analysis		
VOC analysis	sorbent tubes and TD-GC/MS analyses (service)	≈ 500–1 500
surface analysis	SEM/EDX, AFM, Raman, ToF-SIMS/XPS facility time	≈ 1 000–4 000
Total if everything is bought new		**≈ 30 000–50 000**

Using the group's existing optics, power meters, N_2 supply, polishing kit and printers roughly halves the total. If an existing 369.5 nm laser can spare 20 – 40 mW for long runs, the dedicated source (about $8–9k) can be dropped, at the price of competing with the experiments for laser time. The minimum viable thesis (Section 6) needs little beyond test cables, windows, printing materials, gas lines and trace O_2 verification.

7.3 People and facilities

Daily supervision by a group member experienced with UV fibre coupling and polishing. The mechanical workshop for machined sealing lands and possible aluminium cells. Materials-characterisation facilities for surface analysis, contacted in month 2. The departmental laser and gas safety officers for the risk assessments.

8 Risks and mitigation

Table 9: Risk register (L = likelihood, I = impact; H/M/L). To be reviewed at each monthly meeting.

Risk	L	I	Mitigation
Testbed does not reproduce the lab degradation, or it is too slow	M	H	Phase 0 forensics first; increase power or intensity (D1); run one arm next to the experiments to compare environments; treat the difference as a result.
Alignment drift masks end-face changes	H	M	Daily re-peak protocol with logged before/after values; stable mounts; temperature-stable enclosure around the testbed; back-reflection and output-image monitors; sentinel arm.
Detector or reference drift (UV ageing of Si photodiodes, temperature)	M	M	Low intensity on detectors, integrating spheres or diffusers, UV-resistant photodiodes, weekly cross-check with a reference power meter, temperature logging and correction.
Laser instability, mode hops, shutdowns	M	L	Ratio measurement $P_{\text{out}}/P_{\text{ref}}$; uninterruptible supply for laser and logger; automatic restart and alert e-mail.
Cells leak, or printed walls are permeable	M	M	Pressure-decay test of every cell; purge mode as fallback; epoxy sealing; metal or nylon gas lines; continuous O_2 monitoring.
Printed parts, absorber or sealants contaminate the end face	M	H	**N2+P** test arm; low-outgassing materials and bake-out; absorber behind a membrane; no silicone grease; machined aluminium cells as fallback.
Fibre-to-fibre variability (polish, epoxy) larger than the atmosphere effect	M	H	Single batch, one polishing procedure, acceptance criteria, randomised assignment, replicates in Campaign II, mixed-effects analysis.
Lead times or budget for parts	M	M	Order in month 1; borrow from the group's stock; the core arms need little special equipment.
Access to surface-analysis facilities	M	L	Contact facilities in month 2; keep uncleaned ends in clean, N_2-filled containers until analysed.
Negative result: oxygen not the cause	M	L	The design identifies the actual cause (Table 4); this is a valuable result for the group and for publication.
Student overloaded by scope	M	M	Minimum viable thesis defined (Section 6); extended arms optional; weekly priorities agreed with the supervisor.
Safety incident (UV exposure, gas, chemicals)	L	H	Risk assessments before work starts, enclosed beams, interlocks, training (Section 9).

9 Health and safety

Hands-on work starts only after the departmental inductions and with signed risk assessments for the laser work, the gas set-up, and the chemicals (COSHH). The main hazards and controls are:

Near-UV laser radiation
A 375 nm diode of tens of milliwatts, or the 369.5 nm systems, are Class 3B. Near-UV beams are barely visible, so stray reflections are easy to miss. Controls: enclose the beam paths of the testbed (the long runs are unattended), fit an interlock to the enclosure, wear eyewear with adequate optical density at the wavelength in use, use fluorescent viewing cards, register the set-up with the laser safety officer and follow the local rules. Unattended operation needs its own risk assessment.

Compressed and inert gases
Cylinders are secured and fitted with suitable regulators and relief devices. The lab N_2 supply is used where possible. The purge flows are tiny (about 0.1 L/min in total) and cannot deplete the oxygen in a ventilated lab. The scenario that sets the risk is the sudden release of a whole cylinder [143]. A 50 L, 200 bar N_2 cylinder ($\approx$9.5 m^3 of gas) emptied into a 50 m^3 room would lower the oxygen to about 17.6 %, below the 19.5 % minimum, so it would need an oxygen-depletion monitor. A 10 L cylinder would lower it only to about 20.1 %, and it lasts about four months at 10 sccm. Small cylinders are therefore preferred. Exhausts are routed to extraction.

Forming gas
5 % H_2 in N_2 is supplied as a non-flammable compressed-gas mixture; safety data sheets for up to 5.7 % H_2 list no flammability hazard [144]. It is still handled as a hydrogen-containing gas (no ignition sources at the exhaust, leak checks with a hydrogen detector). Use only if Campaign II needs it.

Oxygen absorbers
Iron-based sachets heat up as they react. Keep them sealed until use, do not cut them open, and allow used sachets to finish reacting before disposal.

Chemicals
Isopropanol and acetone (flammable), epoxies and resins (skin sensitisers), polishing slurries. Wear gloves and eye protection and work with ventilation as stated in the COSHH assessments.

3D printing
Run printers in ventilated or filtered enclosures (FDM printers emit ultrafine particles and VOCs). Handle uncured resin with gloves.

Fibre shards
Wear eye protection when cleaving and polishing, use a sharps container and a dedicated work mat, and keep food and drink out of the area.

Plasma and UV/ozone cleaners
If used, they produce UV-C and ozone. Use them only with their interlocks and extraction.

Rule of thumb

If an experiment is going to run unattended for weeks, design it so that every failure (power cut, gas cylinder empty, laser fault, interlock trip) leaves it in a safe state *and* is logged with a timestamp. Then the failure costs you a data gap but does not put anyone at risk.

10 Expected outcomes and impact

For the group. Either a validated, low-cost way to stop the degradation of 369.5 nm fibre links (printed cells with oxygen absorbers or a clean purge), or a clear identification of the cause and a recommendation for fibre or end-face engineering. Both reduce down time, repeated polishing, and slow power and pointing drifts in the experiments. The same cells can protect fibres at 399 nm and other near-UV wavelengths used in the group.

For the field. Fibre delivery of near-UV light is a widespread practical problem in trapped-ion and neutral-atom experiments and in UV photonics, but the role of the atmosphere at the end face has, to our knowledge, not been isolated systematically. A controlled comparison of O_2, water, contaminants and particles at CW near-UV intensities of order 10^5 W/cm^2 is suitable for a short paper (e.g. in *Optics Express*, *Applied Optics*, *Review of Scientific Instruments* or *EPJ Techniques and Instrumentation*). Open CAD files and procedures make the result easy to adopt.

For the student. Training in experimental design for slow, drift-prone measurements, fibre optics and UV laser handling, gas handling, CAD and 3D printing for functional parts, data acquisition and statistics, and surface analysis. This mix of skills is valued in quantum-technology labs and in industry.

Part II – Guide for the student

11 How to use this guide

Part I describes *what* the project should find out and *why*. This part is about *how*: what to do first, which skills to build, how to keep records that will still make sense in month eleven, and which mistakes cost other people months. Treat it as a starting point and improve it. If something here turns out to be wrong for our lab, write that down. Corrections like that belong in your thesis.

Three habits that decide the outcome

1. **Look before you touch.** Image every end face *before* cleaning or polishing it. A cleaned-off film is evidence you can never get back.
2. **Log every intervention.** Every re-peak, swap, cleaning, gas-cylinder change and power cut goes into the event log with a timestamp. Weeks-long runs are only interpretable if you know what happened to them.
3. **Write as you go.** Draft the methods chapter while you build. By month 9 you will have forgotten why you chose a 5 mm lens or a 3 nm filter.

12 Your first four weeks

Table 10: Suggested onboarding plan. Adapt it with your supervisor in the first meeting.

Week	Activities	Output
1	Departmental, laser, gas and chemical inductions; accounts (lab notebook, repository, data storage); read the five starred papers in Section 17; watch an experienced group member polish and inspect a connector	reading notes (one paragraph per paper); induction certificates
2	Practise cleaning, inspection and coupling with a visible laser and a spare fibre; measure coupling efficiency properly (Section 13); collect degraded fibres and their histories from the group	first coupling measurements; inventory of used fibres
3	Phase 0 forensics on the used fibres (imaging, low-power throughput); draft risk assessments; first CAD model and a test print with an O-ring groove; Python logger reading one photodiode	image set; draft risk assessments; logger prototype
4	Write the project plan (2–3 pages: aims, first experiments, parts list with prices and lead times); present it in the group meeting (10 minutes); place orders	MS1: plan and risk assessments signed off, orders placed

13 Core skills

13.1 Handling, cleaning and inspecting fibre end faces

- Never touch an end face, and cap every connector as soon as it is unplugged. Store fibres with caps on, in a clean box.
- Always *inspect first*. If the face is clean, do not clean it. Cleaning tools can leave residues and scratch the face.
- For cleaning, go from dry to wet: filtered gas or a dry cleaning cassette first, then a lint-free wipe moistened with reagent-grade isopropanol, followed immediately by a dry wipe. Inspect again.
- Use the same microscope, magnification, illumination and focus for every image and save the raw files with the fibre ID, end (A/B), date and step in the file name. Use the core and cladding zones of IEC 61300-3-35 as the vocabulary for describing defects.

13.2 Polishing

Learn the group's procedure from an experienced member and then write it down as an SOP with all parameters (films, grit sequence, times, pressure, lubricant, rinsing). Keep a polishing log for every end. Variability between polishes is one of the largest uncontrolled factors in this project. Make polishing boring and reproducible, and measure its outcome (inspection images, coupling efficiency, end-face geometry if an interferometer is available).

13.3 Coupling near-UV light into a single-mode fibre

- **Mode matching.** For a collimated Gaussian beam of $1/e^2$ radius w_{in}, a lens of focal length f produces a waist $w_0 = \lambda f/(\pi w_{\text{in}})$. Choose f so that w_0 equals the fibre mode-field radius. With a mode-field diameter of ≈2.4 µm at 369.5 nm, an $f = 5\,\text{mm}$ asphere needs a ≈1.0 mm diameter beam (Appendix A).
- **Alignment.** Start with a visible alignment beam if the optics allow it. Make the output of the fibre visible, then walk the beam with two mirrors (translation and angle) before touching focus. Optimise the focus last and re-walk.
- **PM fibre.** Align the input polarisation to the slow axis: rotate the input half-wave plate while measuring the output polarisation extinction ratio, and confirm by gently warming or stressing the fibre (the output polarisation should not change).
- **Measure coupling efficiency correctly.** Measure the power directly before the fibre (after all optics, including the cell window) and directly after the output, with the same calibrated sensor. Correct for the Fresnel loss at each face if you report an "internal" efficiency, and state clearly which definition you use.

13.4 Measuring UV power reliably

- Silicon photodiodes slowly lose responsivity under UV exposure, and photodiodes and amplifiers drift with temperature. Keep the intensity on detectors low, prefer integrating spheres or diffusers, and consider photodiodes specified for UV resistance for the long runs. (SiC photodiodes, often recommended for UV, barely respond at 369.5 nm.)
- Check the zero (laser off) and the linearity (calibrated attenuator) at the start of every run.
- Avoid interference fringes: use wedged optics and tilt surfaces slightly, otherwise etalons turn tiny temperature drifts into apparent power changes.
- Cross-check all detectors weekly against one reference power meter that is otherwise kept in a drawer (so it does not age).

13.5 Gas handling and leak testing

- Get trained on regulators and cylinder handling. Use compression fittings correctly (do not over-tighten; replace ferrules) and leak-check every new joint with a leak-detection liquid or a helium or hydrogen sniffer.
- For ppm-level oxygen, the tubing matters. Polyurethane and fluoropolymer tubes are permeable to O_2; stainless steel, copper or short nylon lines are better (Appendix A).
- Leak-test every cell before use: pressurise to about 100 mbar with N_2, log pressure *and* temperature for 12 hours, and compute the leak rate $Q = V\,\mathrm{d}p/\mathrm{d}t$ after correcting for temperature (Appendix B).

13.6 CAD and 3D printing of gas-tight parts

- FDM prints are porous unless printed with 100 % infill, several perimeters, slight over-extrusion and good layer adhesion. Even then, seal the inside with a thin low-outgassing epoxy for the static cells. Resin (SLA) prints are denser but release residual monomer unless fully post-cured. Bake parts (at a temperature safely below the glass transition of the polymer) under N_2 purge before use.

- Design O-ring grooves to standard proportions (squeeze and gland fill as in a sealing handbook) and print the sealing faces flat and smooth, or face them with a thin epoxy layer. Use heat-set metal inserts for screws.
- Choose the material with Table 3 in mind. PETG is a good first choice for FDM. ABS (styrene emission, high water permeability) and standard SLA resins (high outgassing) are poor choices near UV-illuminated end faces. Use walls of at least 1.1 mm and randomised seams [117].
- Print a test coupon with every new material or setting, and keep coupons for the **N2+P** outgassing test.

13.7 Data acquisition and analysis

- Write the logger so that it never loses data. Write to a new file every day, flush after each write, use UTC timestamps, and save a metadata file with fibre IDs, arm assignment, calibration factors and the software version (git commit). Make it restart automatically after a power cut.
- Keep raw data read-only. All corrections (detector temperature, flagged intervals) are applied by analysis scripts, never by editing data files.
- Make every figure in the thesis reproducible from raw data with one script. A notebook is fine for exploration, but the final analysis should be a script in the repository.
- Learn enough statistics to defend an upper limit: regression with autocorrelated residuals, block bootstrap, and mixed-effects models (Section 5.9).

14 Working practices

Lab notebook
Use the group's electronic notebook. Create one entry per day with the plan, what was done, observations, and links to data and images. Note surprises immediately, even if they seem unimportant.

Repository
Keep code, CAD files, SOPs, and the thesis source in version control. Commit small and often with meaningful messages. Store large raw data on the group's storage, not in git, and put the paths in the metadata.

Meetings
Meet the daily supervisor weekly for 30 minutes with a short written agenda (done, problems, plan, decisions needed). Meet the PI monthly, together with a one-page progress report that also reviews the risk register (Table 9). Present in the group meeting about every two months.

Asking for help
If you are stuck for more than a day on something that someone in the group has done before, ask. Experienced colleagues can often solve fibre-coupling and vacuum or gas problems in minutes.

15 Common pitfalls

Lessons from long-term optical measurements

- **Confusing alignment drift with degradation.** Always record throughput before *and* after re-peaking.
- **Cleaning away the evidence.** Image and, if possible, analyse before cleaning.
- **Comparing runs done at different times.** Run conditions in parallel; if you must compare runs, keep a reference arm common to both.
- **Trusting a gas cell you have not measured.** A cell with an indicator that has turned colour, or an O_2 reading that drifts up, means the run is compromised. Log O_2, RH and flow continuously.

- **Introducing contamination with the fix.** Silicone grease, fresh resin prints, adhesive tapes and cable jackets near the end face can all release organics. Keep them out of the cell or test them (**N2+P**).
- **Changing two things at once.** New fibre batch, new polish and new atmosphere in the same run cannot be untangled.
- **Leaving the thesis until the end.** Writing always takes longer than planned, and writing early also shows where data are missing while you can still take them.

16 Writing the thesis

A suggested structure (adapt it to the departmental regulations):

1. Introduction: UV light delivery for trapped ions, the problem, aims.
2. Background: optical fibres and coupling; defects and solarisation in silica; surface photochemistry and laser-induced contamination; gas transport and controlled atmospheres.
3. Methods: testbed, cells and their qualification, procedures, data analysis.
4. Results I: forensics of used fibres and the baseline degradation law.
5. Results II: the atmosphere campaigns and the cleaning sequence; surface analysis.
6. Results III: demonstration on a 369.5 nm link.
7. Discussion: mechanism, comparison with the literature, design guidelines.
8. Conclusions and outlook.

Appendices: CAD drawings and bill of materials, SOPs, logger documentation, data-availability statement. Plan for at least eight weeks of focused writing, and give your supervisor complete chapter drafts early enough for a round of feedback.

17 Reading list

Read the starred items in the first week, the rest as the project needs them. For every paper, note in your reading log the wavelength, intensity, atmosphere, fibre or surface type, time scale and the numbers you might want to compare with later. A one-page table of these numbers will be the backbone of the discussion chapter.

Essential (week 1)

★ Colombe et al. [31]: the reference paper on UV solarisation of single-mode fibres for ion trapping, and on hydrogen loading and curing.

★ Marciniak et al. [46]: UV patch cords in an ion-trap lab, facet deposits and the nitrogen purge. Read the methods carefully: they are close to what you will build.

★ Gonschior et al. [14] and the summary chapters of Gonschior's thesis [49]: the most detailed study of CW blue-light degradation of small-core single-mode fibres, including end caps.

★ Mondin et al. [71] and Hippler et al. [55]: what the space-laser community learned about laser-induced contamination and oxygen.

★ Parthasarathy et al. [74]: CW contamination at an intensity similar to ours, as a function of oxygen pressure.

Background for the thesis (weeks 2–8)

- Defects in silica: Skuja [4], Skuja et al. [6], Griscom [5] and the fibre-oriented review by Girard et al. [8]. Oxygen species in silica: Kajihara et al. [145].

- Surfaces: Radzig [43] on activated silica surfaces.
- Fibre alternatives: Gebert et al. [47] and Dorer et al. [61] (hollow core), Yamamoto et al. [54] (LMA PCF in the deep UV).
- Oxygen in UV laser systems: Wilson et al. [75] and Hannig et al. [76] (UV conversion crystals in ion-trap labs), Schoedl et al. [81] (laser-diode facets in nitrogen versus oxygen), and the Fujifilm patent on siloxane deposits at fibre end faces [89].
- Optics: Kogelnik and Li [146] on Gaussian beams (mode matching), Lax [147] on laser heating.
- Cleaning: Vig [93] on UV/ozone cleaning.

Practical references

IEC 61300-3-35 for end-face inspection [140]; Thorlabs' polishing instructions and fibre damage-threshold tutorial [59, 141]; the NIST UV patch-cord recipe [60]; photodiode data sheets with notes on UV exposure [139, 142]; and the references on 3D-printed vacuum and gas parts in Section 3.5.

Appendices

A Design estimates

All numbers in this appendix are printed by `calc/estimates.py` in the repository, where every input is documented. The student should rerun it with measured values (e.g. the actual mode-field diameter and cell volume) and keep it up to date.

A1 – Intensity at the end face. For a Gaussian mode of $1/e^2$ radius w the peak intensity is $I_0 = 2P/(\pi w^2)$. With a mode-field diameter of 2.4 µm at 369.5 nm:

Guided power	1 mW	3 mW	10 mW
peak intensity I_0 (kW/cm^2)	44	133	442
photon flux (cm^{-2} s^{-1})	8×10^{22}	2.5×10^{23}	8×10^{23}
fluence per day (J/cm^2)	3.8×10^{9}	1.1×10^{10}	3.8×10^{10}

A2 – How much contamination is "a lot"?. From kinetic theory, a molecule of mass 150 u present at 1 ppb strikes the surface at 1.3×10^{18} m^{-2} s^{-1}. That is a monolayer-equivalent every ∼8 s. If only one impact in a million is photo-fixed, a monolayer still grows every ∼90 days at 1 ppb, every ∼9 days at 10 ppb and about daily at 100 ppb. Parts-per-billion levels of organics are therefore enough to explain degradation over days to weeks. This is consistent with the Aeolus experience [66], and it is why the purge gas needs a purifier and the cells must be clean.

A3 – How thick must a film be to matter?. An absorbing film with extinction coefficient k absorbs 10 % at a thickness of 103 nm, 31 nm and 10 nm for $k = 0.03$, 0.1 and 0.3 (polymer-like to graphitic carbon; the values of k are assumptions). A transparent film ($n = 1.5$) whose thickness follows the Gaussian intensity profile reduces the coupling efficiency as $\eta = \mathrm{sinc}^2(\phi_0/2)$, where ϕ_0 is the peak phase. A 10 % loss needs $\phi_0 = 1.1\,\mathrm{rad}$, i.e. a peak thickness of ≈130 nm. Absorbing films of a few tens of nanometres, or transparent ones of about a hundred, are therefore enough. Both are easy to see by AFM or white-light interferometry.

A4 – Heating by an absorbing film. For absorption at the surface of a semi-infinite silica body, the steady-state temperature rise at the centre is $\Delta T = P_{\mathrm{abs}}/(\sqrt{2\pi}\,\kappa w)$ with $\kappa = 1.38\,\mathrm{W/(m\,K)}$ [147]. A film that absorbs 10 % of 3 mW heats the face by ≈70 K, and at 10 mW by ≈240 K. Once a film absorbs, it heats itself and its surroundings. This feedback can accelerate its own growth and explains why degradation often appears to "run away".

A5 – Gas transport into silica. With literature Arrhenius parameters at 295 K:

	D (cm^2/s)	$\sqrt{Dt}$, 1 year	time constant, 125 µm fibre
H_2 [37, 38]	0.9×10^{-11} – 2.2×10^{-11}	170 – 260 µm	3.6 – 8.8 days
O_2 [1, 39]	3×10^{-24} – 6×10^{-21}	0.1 – 4 nm	$> 10^{10}$ days

The oxygen values are extrapolated from measurements above 800 °C and are good to an order of magnitude at best. The conclusion is robust: oxygen reaches only the top few nanometres of the end face, whereas hydrogen penetrates the whole fibre within days.

A6 – Continuous purge. A well-mixed cell of volume V purged with Q approaches the feed composition with time constant V/Q. For $V = 30\,\mathrm{cm}^3$ and $Q = 20\,\mathrm{sccm}$ (time constant 1.5 min) the ideal time to reach 10 ppm is 15 min. Allowing a factor of three for imperfect mixing, start measurements no earlier than an hour after closing the cell. The steady-state O_2 fraction is set by the influx divided by Q:

- permeation through dense printed walls (60 cm^2, 3 mm; PET 0.034 Barrer to PP 1.7 Barrer, Table 3) adds only 0.1 – 6.5 ppm at 5 sccm;
- one metre of 6 × 4 mm polyurethane, PTFE or FEP tubing adds 70 – 130 ppm at 5 sccm, nylon ∼4 ppm, and stainless steel nothing. *Tubing, not the printed cell, limits the purity.*
- Water is a different matter. At 50 % lab RH, 0.2 mg/day of water passes the same wall in PET or PEEK, but 3.8 mg/day in PLA or ABS (about 650 ppmv at 5 sccm).
- Back-diffusion through a 30 cm long, 2 mm bore outlet is suppressed by $\exp(-\mathrm{Pe})$, with Péclet number $\mathrm{Pe} = QL/(AD) \approx 40$ even at 0.5 sccm ($D_{O_2\text{–}N_2} = 0.20\,\mathrm{cm^2/s}$ [148]). A modest outlet line is therefore as effective as a bubbler, provided the flow never stops.

A7 – Static absorber cell. A sealed 30 cm^3 cell of air contains 6.3 cm^3 of O_2. Absorbing it lowers the pressure of a rigid cell by ≈210 mbar, which then sucks air in through any leak. After an N_2 pre-purge to 1 % O_2 only 0.3 cm^3 remains and the pressure drop is ≈10 mbar. The absorber then has to cope mainly with leaks and permeation. A ZP-50-size sachet (50 cm^3 of O_2) lasts about 8 years at an air leak rate of 1 × 10^{-6} mbar L/s, about 9 months at 1 × 10^{-5} mbar L/s and only about 4 weeks at 1 × 10^{-4} mbar L/s. Hence the leak-rate specification of 1 × 10^{-5} mbar L/s in Section 5.6. Permeation adds to the load: a 2 mm FKM or butyl O-ring of 30 mm diameter lets in about 2 µL of O_2 per day (65 ppm/day in a sealed 30 cm^3 cell without absorber), a silicone O-ring about 200 times more. With an absorber, the equilibrium concentration in a leaking box is $[O_2] \approx L/(12.7\,C)$ [102]. An ingress of 200 ppm/day and $C = 50/30$ give about 10 ppm, comfortably below the 100 ppm target, as long as the absorber is not exhausted. Water ingress, by contrast, saturates an undesiccated cell within hours (A6). A pressure-decay test with a 0.1 mbar sensor over 12 h resolves ∼1 × 10^{-7} mbar L/s. A temperature change of only 0.1 K shifts the pressure by 0.34 mbar, so temperature must be logged and corrected for.

A8 – Optics. Mode matching to a 2.4 µm mode needs collimated beam diameters of 0.88 mm, 0.98 mm and 1.57 mm for $f = 4.5\,\mathrm{mm}$, 5 mm and 8 mm ($w_{\mathrm{in}} = \lambda f/\pi w_0$ [146]). A 1 mm fused-silica window ($n = 1.474$ [149]) in a beam converging at NA 0.1 adds 5 nm of primary spherical aberration (Strehl ratio > 0.9999) and moves the focus by 0.32 mm. A bare end face reflects 3.7 %. At an 8° APC face the input beam must arrive 3.8° off the fibre axis, and the reflection leaves 23.7° away from the incoming beam. A coreless end cap of length L expands the beam before the glass–gas interface (Rayleigh range in glass 18 µm). This reduces the face intensity 32-fold for $L = 100\,\mu\mathrm{m}$, 124-fold for 200 µm, 277-fold for 300 µm and 770-fold for 500 µm.

A9 – How small a degradation rate can be demonstrated?. For N daily readings over T days with scatter σ, the standard error of a fitted slope is $\sigma/(\sqrt{N}\,T/\sqrt{12})$. With $\sigma = 0.3\,\%$, three weeks give a 95 % upper limit of ≈0.9 % per 1000 h, and six weeks give ≈0.3 % per 1000 h. The "degradation halted" criterion of Section 5.9 ($< 1\,\%$ per 1000 h) is therefore within reach of a single long run, provided systematic drifts are controlled to the same level. Controlling them is the real challenge.

B Draft standard operating procedures

These drafts are a starting point. The student turns them into the group's SOPs during WP0–WP3, with the actual equipment, part numbers and parameters.

SOP-1: End-face inspection and cleaning

1. Wear gloves; work on a clean mat; have caps ready.
2. Inspect the end face with the inspection microscope at the standard settings. Save the image as `<fibreID>_<end>_<YYYYMMDD>_<step>.png`.
3. If clean (no defects in the core zone, nothing loose in the cladding zone): stop and cap.
4. Dry clean (filtered gas, then a dry cleaning cassette or click cleaner); inspect and image.

5. If still contaminated: wet–dry clean with reagent-grade isopropanol on a lint-free wipe, then immediately a dry wipe; inspect and image. Repeat at most twice.
6. Record every step in the fibre's log. Never clean a fibre that is scheduled for surface analysis.

SOP-2: Polishing and acceptance of a test end

1. Polish with the group's procedure (films, sequence, times, pressure, lubricant written down) and log all parameters.
2. Clean and inspect (SOP-1). Measure end-face geometry if an interferometer is available.
3. Measure the coupling efficiency at low power ($\leq$0.3 mW) with the standard launch.
4. Accept if the core zone is defect-free and the efficiency is within the agreed band around the batch median; otherwise re-polish. Cap and store the fibre in a clean container until use.

SOP-3: Cell qualification (leak and oxygen hold test)

1. Clean all parts (isopropanol wipe, no silicone grease), bake printed parts under N_2 purge, assemble with a dummy connector.
2. Pressure decay: pressurise to 100 mbar with N_2, close the valves, log pressure and temperature for 12 h. Correct for temperature ($\Delta p/p = \Delta T/T$) and compute $Q = V\,\mathrm{d}p/\mathrm{d}t$. Accept if $Q < 1 \times 10^{-5}\,\mathrm{mbar\,L/s}$.
3. Oxygen hold (static cells): purge with N_2 to $\lesssim 1\,\%$ O_2, insert the absorber, seal, and log the O_2 sensor spot reading for at least 7 days. Accept if O_2 stays below 100 ppm after the initial absorption period.
4. Record the cell ID, material, print settings, sealing and test results in the cell log.

SOP-4: Start of a run

1. Fibres accepted (SOP-2), IDs and ends assigned to arms by randomisation, assignment saved in the run metadata.
2. Cells qualified (SOP-3); purge flows set and logged; O_2, RH and VOC readings stable.
3. Detectors zeroed; reference power meter cross-check recorded.
4. Couple each arm at low power, then raise to the test power and set equal powers; record the initial values.
5. Start the logger; write the run plan (duration, re-peak schedule, stop criteria) into the notebook.

SOP-5: End of a run

1. Record the final throughput before and after re-peaking at the test power, then at low power.
2. Switch off the laser. Inspect both ends without touching (SOP-1, imaging only).
3. Either run the cleaning sequence of Section 5.7.1 (output end first) with a low-power measurement after each step, or cap the fibre and store it in an N_2-filled, labelled container for surface analysis.
4. Archive the data, images and notebook entries of the run and write a one-page run summary.

C Data-logging specification

The run metadata (JSON or YAML) contains: run ID, start and end time, fibre IDs and ends per arm, atmosphere per arm, gas cylinder IDs and certificates, cell IDs, test power, detector calibration factors, software version (git commit) and the analysis plan version.

Table 11: Minimum channel list for the logger. Raw sampling at 1 Hz; stored as 1-minute means with standard deviations; one file per day plus a metadata file per run.

Channel	Description	Units
`t_utc`	timestamp (ISO 8601, UTC)	–
`arm<k>_ref`, `arm<k>_out`, `arm<k>_back`	photodiode voltages of arm k	V
`arm<k>_O2`, `arm<k>_RH`, `arm<k>_VOC`	atmosphere of arm k (cell or outlet)	ppm, %, index
`arm<k>_flow`, `arm<k>_T`	purge flow, temperature at the cell	sccm, °C
`laser_I`, `laser_T`, `laser_P`	diode current, temperature, monitor	mA, °C, V
`lab_T`, `lab_RH`, `lab_p`	lab environment	°C, %, hPa
`event`	free-text event log entries (re-peak, swap, cleaning, outage)	–

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